



Verifying the Upper Confidence Bound Algorithm in Isabelle/HOL

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Abstract

This project formally verifies the Upper Confidence Bound (UCB) algorithm in Isabelle/Higher-order Logic (HOL), focusing on its probabilistic guarantees and regret bounds. By leveraging advanced stochastic tools such as martingales, concentration inequalities, and stopping times. This project develops machine-checked proofs of UCB's correctness. The work extends Isabelle/HOL's probabilistic framework and explores verification of continuous-time bandit models via stochastic differential equations and Itô calculus. This research advances the formal verification of probabilistic algorithms in reinforcement learning.

I declare that this dissertation was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified. The research complies with academic integrity standards.

(Arjan Faber)

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1 Introduction

The Upper Confidence Bound (UCB) algorithm stands as a theoretically grounded, asymptotically optimal strategy for addressing the exploration-exploitation trade-off in stochastic multi-armed bandit problems that underpins a wide array of high-impact applications from adaptive recommendation systems to real-time decision-making in dynamic environments; as such, early and formal verification of its regret guarantees is not merely academic, but foundational to the trustworthiness and reliability of modern online learning frameworks.

The increasing complexity of decision-making algorithms in reinforcement learning (RL) and bandit problems calls for rigorous verification. The Upper Confidence Bound (UCB) algorithm is a cornerstone in stochastic decision-making, yet its theoretical guarantees often hinge on heuristic reasoning. Formal verification provides a robust approach to proving properties such as regret bounds and convergence guarantees.

Isabelle/HOL, a powerful interactive theorem prover, supports the formalization of measure-theoretic probability and stochastic processes. While foundational results in probability theory and basic reinforcement learning have been formalized, a complete machine-checked proof of the classical UCB algorithm remains unachieved. This dissertation addresses that gap by formalizing key mathematical tools required for UCB’s theoretical analysis. Martingales are used to model the reward estimation process, allowing us to formally prove that the empirical mean converges to the true mean over time. Concentration inequalities, such as Hoeffding’s, are essential for quantifying the probability that the empirical mean significantly deviates from the expected value—an important step in bounding regret. Stopping time theorems, particularly the Optional Stopping Theorem (OST), are necessary to handle adaptive decision-making, where the number of samples drawn from each arm depends on the history of observed outcomes. By formalizing these tools in Isabelle, the project provides the foundation for verifying the regret guarantees of both discrete and continuous-time versions of UCB.

This dissertation leverages Isabelle/HOL to formally verify UCB, providing machine-checked proofs of regret guarantees and extending Isabelle’s probability libraries. The outcomes support robust decision-making in AI, finance, healthcare, and autonomous systems.

1.1 Problem Statement

Despite UCB’s theoretical significance, its regret guarantees have not been formally verified in theorem provers. This dissertation addresses this gap by extending Isabelle’s probability theory library with martingales, stopping times, and concentration inequalities. Also, we want to formalize regret bounds and proving UCB’s correctness. Finally, our goal is to explore continuous-time UCB via stochastic differential equations (SDEs). Key guiding research questions for our project include:

1. How can measure-theoretic probability be effectively represented in Isabelle?
2. How can we formally prove concentration bounds and regret guarantees?

3. How can martingale tools like OST and MCLT be formalized and applied?
4. How can continuous-time bandit processes be captured using Isabelle?

2 Literature Review

The formal verification of probabilistic and learning systems has been an active area of research, leveraging rigorous mathematical foundations to ensure reliability and robustness.

Classical stochastic processes and martingales provide the theoretical backbone for many modern analyses. Foundational works such as Doob (1953) establish the mathematical framework of continuous martingales and Brownian motion, critical for modelling uncertainty in dynamic systems. Dudley’s seminal work Dudley (1967) further elucidates the structure of Gaussian processes, essential for understanding function spaces underpinning probabilistic models.

Within machine learning, multi-armed bandit (MAB) problems represent a fundamental paradigm for sequential decision-making under uncertainty. A finite-time analysis (Auer et al. (2002)) introduced the widely adopted UCB algorithm, providing regret guarantees pivotal for balancing exploration and exploitation.

3 Bandit Regret: Preliminarily Result

3.1 Isabelle Files

We provide a rigorous description of all parameters and variables involved in the formalization of the multi-armed bandit problem and policy within Isabelle/HOL, specifying their respective domains, types, and mathematical properties. I refer to Isabelle (.thy) files for the verification of theorem 15.4 and intermediate results proposition 15.1, proposition 15.5 and lemma 15.6 all from Rebeschini (2022) .

3.2 Mathematical Explanation

Let $A \subseteq \mathcal{A}$ be a finite set representing the available arms, where $\mathcal{A}$ is the universal set of arms (corresponding to the Isabelle type `'a`). Formally, A belongs to the finite powerset of $\mathcal{A}$, denoted by $\mathcal{P}_{\text{fin}}(\mathcal{A})$, and satisfies $|A| < \infty$. This finiteness assumption ensures that all sums indexed over A are well-defined and finite.

The expected reward function $\mu : A \rightarrow \mathbb{R}$ assigns a deterministic real value to each arm $a \in A$. The function space $\mu \in \mathbb{R}^A$ consists of all real-valued functions on A . An optimal arm $a_\star \in A$ satisfies the maximality condition

$$\forall a \in A, \quad \mu(a_\star) \geq \mu(a).$$

Accordingly, we define the sub-optimality gap function $\Delta : A \rightarrow \mathbb{R}_{\geq 0}$ by

$$\Delta(a) := \mu(a_\star) - \mu(a),$$

Parameter	Isabelle Type	Mathematical Domain
Arm set	<code>'a set</code>	finite subset $A \subseteq \mathcal{A}$
Expected reward	<code>'a $\Rightarrow$ real</code>	$\mu : A \rightarrow \mathbb{R}$
Optimal arm	<code>'a</code>	$a_* \in A$
Gap function	<code>'a $\Rightarrow$ real</code>	$\Delta : A \rightarrow \mathbb{R}_{\geq 0}$
Sample space	<code>'b set</code>	Ω
σ -algebra	<code>'b set set</code>	$\mathcal{F} \subseteq \mathcal{P}(\Omega)$
Probability measure	<code>'b measure</code>	$M : \mathcal{F} \rightarrow [0, 1]$
Policy	<code>nat $\Rightarrow$ 'b $\Rightarrow$ 'a</code>	$\pi : \mathbb{N} \times \Omega \rightarrow A$
Pull counts	<code>nat $\Rightarrow$ 'a $\Rightarrow$ 'b $\Rightarrow$ nat</code>	$N_n : \mathbb{N} \times A \times \Omega \rightarrow \mathbb{N}_0$

Table 1: Problem definition

which is always nonnegative, with $\Delta(a_*) = 0$.

Next, the policy framework is defined over a probability space $(\Omega, \mathcal{F}, P)$, where Ω is a nonempty sample space (Isabelle type `'b set`), $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra of measurable subsets, and $P : \mathcal{F} \rightarrow [0, 1]$ is a probability measure such that $P(\Omega) = 1$.

A stochastic policy $\pi : \mathbb{N} \times \Omega \rightarrow A$ assigns at each discrete time step $t \in \mathbb{N}$ and for each outcome $\omega \in \Omega$ an arm $\pi(t, \omega) \in A$. The policy satisfies the measurability condition that for fixed t , the mapping $\pi(t, \cdot) : (\Omega, \mathcal{F}) \rightarrow (A, \mathcal{P}(A))$ is measurable with respect to the counting σ -algebra on A .

For each time $n \in \mathbb{N}$, arm $a \in A$, and sample path $\omega \in \Omega$, define the arm pull count

$$N_n(n, a, \omega) = |\{t \in \{0, \dots, n-1\} \mid \pi(t, \omega) = a\}|,$$

which counts how many times arm a was selected up to time n . The function N_n is measurable in ω for fixed n and a .

Since exactly one arm is pulled at each time step, the sum over all arms satisfies

$$\sum_{a \in A} N_n(n, a, \omega) = n,$$

for every $n \in \mathbb{N}$ and $\omega \in \Omega$.

A summary table of the parameters, their Isabelle types, and their mathematical domains follows: These definitions provide the foundational mathematical structure for subsequent formalizations and proofs related to regret analysis in the stochastic multi-armed bandit framework.

The following Isabelle theories are central to the formalization:

- `HOL-Probability.Probability`
- `HOL-Library.Finite_Map`

- HOL-Analysis.Analysis
- HOL-Analysis.Bochner_Integration

3.3 Locale Setup: Bandit Problem Definition

This section presents a formalization of the regret in stochastic multi-armed bandit problems using Isabelle/HOL. Each Isabelle code fragment is paired with its corresponding mathematical formulation and interpretation.

3.3.1 Parameter and Variable Definitions

Let A be a finite set representing the available arms. For each arm $a \in A$, we define a mean reward function $\mu : A \rightarrow \mathbb{R}$. The optimal arm $a_\star \in A$ satisfies the maximality condition

$$\mu(a_\star) \geq \mu(a) \quad \text{for all } a \in A.$$

We define the sub-optimality gap function $\Delta : A \rightarrow \mathbb{R}_{\geq 0}$ as

$$\Delta(a) := \mu(a_\star) - \mu(a),$$

which quantifies the loss incurred by selecting arm a instead of the optimal arm $a_\star$. Clearly, $\Delta(a_\star) = 0$ and $\Delta(a) \geq 0$ for all $a \in A$. Section 10.1 in the appendix describes the problem definition for the proposition 15.1 in Isabelle.

3.4 Regret Decomposition (Pointwise)

Let $N_n(n, a, \omega)$ denote the number of times arm a has been pulled up to time n , under outcome ω . We assume that the total number of arm selections equals n , i.e.,

$$\sum_{a \in A} N_n(n, a, \omega) = n.$$

The cumulative regret at time n , under sample path ω , is defined by

$$R_n(n, \omega) = n \cdot \mu(a_\star) - \sum_{a \in A} \mu(a) \cdot N_n(n, a, \omega).$$

We manipulate this expression algebraically:

$$\begin{aligned} R_n(n, \omega) &= \sum_{a \in A} \mu(a_\star) \cdot N_n(n, a, \omega) - \sum_{a \in A} \mu(a) \cdot N_n(n, a, \omega) \\ &= \sum_{a \in A} (\mu(a_\star) - \mu(a)) \cdot N_n(n, a, \omega) \\ &= \sum_{a \in A} \Delta(a) \cdot N_n(n, a, \omega). \end{aligned}$$

Thus, the regret decomposes as a weighted sum of the number of times each arm is pulled, where the weights are the sub-optimality gaps.

In Isabelle we implemented this as follows:

```

lemma regret_decomposition_pointwise:
  fixes n :: nat and  $\omega$  :: 'b
  assumes n_count_assm_pointwise: "(\sum a \in A. real (N_n n a  $\omega$ )) = real n"
  shows "R_n n  $\omega$  = (\sum a \in A.  $\Delta$  a * real (N_n n a  $\omega$ ))"
proof - [23 lines]
qed

```

Figure 1: Regret decomposition (point-wise) (*MSc_Project_Discrete_Prob15-1.thy*)

3.5 Expected Regret Decomposition

In Isabelle we know that we can use the **HOL-Probability.Probability** library that supports:

Definition *expectation* : $(\Omega \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ where $\mathbb{E}(X) := \int_{\Omega} X(\omega) dP(\omega)$,

where P is a probability measure defined on the measurable space $(\Omega, \mathcal{F})$, and the integral is taken in the Bochner sense.¹²

We assume that the action set A is finite and that for each $a \in A$, the random variable $\omega \mapsto N_n(n, a, \omega)$ is integrable under the probability measure P , i.e.,

$$\forall a \in A, \quad N_n(n, a, \cdot) \in L^1(P).$$

First, we observe that the function $\omega \mapsto n \cdot \mu(a_*)$ is constant and thus integrable.

Next, we note that each term $\omega \mapsto \mu(a) \cdot N_n(n, a, \omega)$ is the product of a constant and an integrable function, hence integrable. By the finiteness of A , the sum

$$\omega \mapsto \sum_{a \in A} \mu(a) \cdot N_n(n, a, \omega),$$

is also integrable.

The third step invokes the pointwise decomposition established earlier, expressing regret as

$$R_n(n, \omega) = \sum_{a \in A} \Delta(a) \cdot N_n(n, a, \omega).$$

Taking expectations on both sides yields

$$\mathbb{E}[R_n(n)] = \mathbb{E} \left[\sum_{a \in A} \Delta(a) \cdot N_n(n, a) \right].$$

¹We use here the mathematical notation of the expectation using *expectation*($\cdot$) = $\mathbb{E}(\cdot)$

²The **Bochner integral** extends the Lebesgue integral to functions taking values in a Banach space B , such as $\mathbb{R}^n$ or function spaces. Specifically, it allows integration of measurable functions $X : \Omega \rightarrow B$ where B is a complete normed vector space. The Bochner integral reduces to the Lebesgue integral when $B = \mathbb{R}$.

In Isabelle/HOL, the **HOL-Analysis.BochnerIntegration** library provides the framework for defining integrals of vector-valued functions, while **HOL-Probability.Probability** uses the Bochner integral to define expectation more generally.

Due to linearity of expectation and the finiteness of A , we are permitted to exchange the sum and expectation:

$$\mathbb{E}[R_n(n)] = \sum_{a \in A} \Delta(a) \cdot \mathbb{E}[N_n(n, a)].$$

This completes the proof of the expected regret decomposition:

$$\mathbb{E}[R_n(n)] = \sum_{a \in A} \Delta(a) \cdot \mathbb{E}[N_n(n, a)].$$

4 Step 1: Discrete UCB algorithm Proof Verification

This section guides through the first step of the discrete UCB proof of Rebeschini (2022) using our definitions and results from proposition 15.1.

4.1 Lemma union_eq

The lemma `union_eq` establishes a fundamental property about partitioning a set of indices based on a threshold k . Specifically, it states:

$$\{t \mid t < n \wedge \pi(t, \omega) = a\} = \{t \mid t < k \wedge \pi(t, \omega) = a\} \cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\},$$

where $k \leq n$, a is a fixed action, and π is the policy selecting actions based on time t and outcome ω . In Isabelle, the lemma is stated as:

```
lemma union_eq:
  fixes a :: 'a and n k :: nat and ω :: 'ω
  assumes "k ≤ n"
  shows "{t. t < n ∧ π t ω = a} = {t. t < k ∧ π t ω = a} ∪ {t. k ≤ t ∧ t < n ∧ π t ω = a}"
proof [8 lines]
next [7 lines]
qed
```

Figure 2: Union equality (*Discrete_UCB_Step1.thy*)

4.1.1 Detailed Proof Explanation

We prove this equality by showing the two subset relations that define set equality:

$$S_1 = \{t \mid t < n \wedge \pi(t, \omega) = a\}.$$

$$S_2 = \{t \mid t < k \wedge \pi(t, \omega) = a\} \cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\}.$$

Our goal is to show $S_1 \subseteq S_2$ and $S_2 \subseteq S_1$.

First Inclusion: $S_1 \subseteq S_2$

We start by showing

$$\{t \mid t < n \wedge \pi(t, \omega) = a\} \subseteq \{t \mid t < k \wedge \pi(t, \omega) = a\} \cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\}.$$

```
show "{t. t < n ∧ π t ω = a} ⊆ {t. t < k ∧ π t ω = a} ∪ {t. k ≤ t ∧ t < n ∧ π t ω = a}"
proof
  fix x assume "x ∈ {t. t < n ∧ π t ω = a}"
  then have "x < n" and "π x ω = a" by auto
  then have "x < k ∨ k ≤ x" by auto
  thus "x ∈ {t. t < k ∧ π t ω = a} ∪ {t. k ≤ t ∧ t < n ∧ π t ω = a}"
    using <x < n> <π x ω = a> by auto
qed
```

Figure 3: Proof of Union_Eq lemma (part i) (*Discrete_UCB_Step1.thy*)

Explanation: We consider an arbitrary element x in S_1 . By the definition of set membership,

$$x \in S_1 \implies x < n \quad \text{and} \quad \pi(x, \omega) = a.$$

Since $k \leq n$, any number $x < n$ must either satisfy

$$x < k \quad \text{or} \quad k \leq x.$$

This case distinction partitions the timeline at k .

- *Case 1:* If $x < k$, then x belongs to the first set on the right side:

$$x \in \{t \mid t < k \wedge \pi(t, \omega) = a\}.$$

- *Case 2:* Otherwise, if $k \leq x < n$, then x belongs to the second set on the right side:

$$x \in \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\}.$$

In either case, x belongs to the union:

$$x \in S_2 = \{t \mid t < k \wedge \pi(t, \omega) = a\} \cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\}.$$

This concludes the proof that any $x \in S_1$ lies in S_2 .

4.1.2 Second Inclusion: $S_2 \subseteq S_1$

Now, we prove the converse:

$$\{t \mid t < k \wedge \pi(t, \omega) = a\} \cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\} \subseteq \{t \mid t < n \wedge \pi(t, \omega) = a\}.$$

```

next
show "{t. t < k ∧ π t ω = a} ∪ {t. k ≤ t ∧ t < n ∧ π t ω = a} ⊆ {t. t < n ∧ π t ω = a}"
proof
  fix x assume "x ∈ {t. t < k ∧ π t ω = a} ∪ {t. k ≤ t ∧ t < n ∧ π t ω = a}"
  then have "x < k ∧ π x ω = a ∨ (k ≤ x ∧ x < n ∧ π x ω = a)" by auto
  hence "x < n ∧ π x ω = a" using assms by auto
  thus "x ∈ {t. t < n ∧ π t ω = a}" by auto
qed

```

Figure 4: Proof of Union_Eq lemma (part ii) (*Discrete_UCB_Step1.thy*)

We take an arbitrary element $x \in S_2$, which means x belongs to either the first set: $x < k$ and $\pi(x, \omega) = a$, or the second set: $k \leq x < n$ and $\pi(x, \omega) = a$.

Since $k \leq n$, in both cases it follows that $x < n$ and $\pi(x, \omega) = a$, which places x in S_1 :

$$x \in \{t \mid t < n \wedge \pi(t, \omega) = a\}.$$

This concludes the proof of the second inclusion and completes the lemma.

The lemma `union_eq` confirms the intuitive fact that splitting the index set $\{0, \dots, n-1\}$ at any point $k \leq n$ partitions the times where the policy selects action a into two disjoint subsets: those before k , and those from k up to $n-1$.

4.2 Cardinality and Indicator Function Equality

In this section, we prove the following key lemma that connects the cardinality of a set defined by a predicate with the summation of indicator functions over a range of natural numbers.

Let $I : \mathbb{N} \rightarrow \{\text{True}, \text{False}\}$ be a boolean predicate. For natural numbers $k \leq n$ and fixed ω , a , the following holds:

$$\text{card}\{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge I(t)\} = \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega) = a \wedge I(t)}.$$

where $\mathbf{1}_P$ is the indicator function equal to 1 if predicate P is true, and 0 otherwise.

```

lemma cardinality_indic_eq:
  fixes I :: "nat ⇒ bool"
  assumes "finite {t. k ≤ t ∧ t < n}"
  shows "card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ I t} = (∑ t = k..<n. if π t ω = a ∧
I t then 1 else 0)"
proof - [38 lines]
qed

```

Figure 5: Lemma cardinality of indicators (*Discrete_UCB_Step1.thy*)

4.3 Step-by-step Proof Explanation

The first step is to establish the finiteness of the underlying set over which we sum or count. The lemma assumes finiteness explicitly:

$$\text{finite}\{t \mid k \leq t < n\},$$

which is straightforward because this is a finite interval of natural numbers. We record this as:

```

have fin: "finite {t. k ≤ t ∧ t < n}" using assms by simp

```

Figure 6: Proof of finite set (*Discrete_UCB_Step1.thy*)

The set $\{t \mid k \leq t < n\}$ corresponds exactly to the integers from k to $n - 1$, which is finite since $k \leq n$. This allows us to use properties of sums and cardinalities over finite sets.

Next, we use a standard lemma in Isabelle called ‘sum_indicator_eq_card’. This lemma states that the sum of indicator functions over a finite set equals the cardinality of that set intersected with the indicator predicate.

Formally:

$$\sum_{t \in A} \mathbf{1}_{P(t)} = \text{card}(A \cap \{t \mid P(t)\}).$$

Applied here with

$$A = \{t \mid k \leq t < n\} \quad \text{and} \quad P(t) \equiv (\pi(t, \omega) = a \wedge I(t)).$$

we get:

```

have sum_eq:
  "sum (indicator {t.  $\pi$  t  $\omega$  = a  $\wedge$  I t}) {t. k  $\leq$  t  $\wedge$  t < n} =
  card ({t. k  $\leq$  t  $\wedge$  t < n}  $\cap$  {t.  $\pi$  t  $\omega$  = a  $\wedge$  I t})"
by (rule sum_indicator_eq_card[OF fin])

```

Figure 7: Sum of indicators (*Discrete_UCB_Step1.thy*)

We chose to define the indicator function I and connect it later on to the in-built indicator function in the proof only because the library lemmas are written that way, but the final statement of our lemma is expressed in terms of I , which is the actual condition we care about. The indicator function $\mathbf{1}_P$ corresponds in Isabelle to ‘indicator t. P t’, which is 1 if t is in the predicate set, 0 otherwise. Since A is finite, the lemma applies directly.

We note that

$$\{t \mid k \leq t < n\} \cap \{t \mid \pi(t, \omega) = a \wedge I(t)\} = \{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge I(t)\}.$$

This is by simple set intersection properties, reflected in Isabelle as:

```

have sets_eq:
  "{t. k  $\leq$  t  $\wedge$  t < n}  $\cap$  {t.  $\pi$  t  $\omega$  = a  $\wedge$  I t} = {t. k  $\leq$  t  $\wedge$  t < n  $\wedge$   $\pi$  t  $\omega$  = a  $\wedge$  I t}"
by auto

```

Figure 8: Set intersection (*Discrete_UCB_Step1.thy*)

From this equality of sets, their cardinalities are equal: We obtain:

```

hence card_eq:
  "card ({t. k ≤ t ∧ t < n} ∩ {t. π t ω = a ∧ I t}) =
card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ I t}"
  by simp
have card_sum_eq:
  "sum (indicator {t. π t ω = a ∧ I t}) {t. k ≤ t ∧ t < n} =
card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ I t}"
  using sum_eq sets_eq by simp
have ind_eq:
  "sum (indicator {t. π t ω = a ∧ I t}) {t. k ≤ t ∧ t < n} =
sum (λt. of_bool (π t ω = a ∧ I t)) {t. k ≤ t ∧ t < n}"
  by (simp add: indicator_def)
have bool_if_eq:
  "sum (λt. of_bool (π t ω = a ∧ I t)) {t. k ≤ t ∧ t < n} =
sum (λt. if π t ω = a ∧ I t then 1 else 0) {t. k ≤ t ∧ t < n}"
  by (simp add: of_bool_def)
have set_eq: "{t. k ≤ t ∧ t < n} = {k..} ∩ {..<n}" by auto
have sum_set_eq:
  "sum (λt. if π t ω = a ∧ I t then 1 else 0) {t. k ≤ t ∧ t < n} =
sum (λt. if π t ω = a ∧ I t then 1 else 0) ({k..} ∩ {..<n})"
  by (simp add: set_eq)
have atLeastLessThan_eq:
  "sum (λt. if π t ω = a ∧ I t then 1 else 0) ({k..} ∩ {..<n}) =
sum (λt. if π t ω = a ∧ I t then 1 else 0) (atLeastLessThan k n)"
  by (simp add: atLeastLessThan_def)
have final_sum_eq:
  "sum (λt. if π t ω = a ∧ I t then 1 else 0) (atLeastLessThan k n) =
∑ t = k..<n. if π t ω = a ∧ I t then 1 else 0"
  by simp
show ?thesis [8 lines]
qed

```

Figure 9: Proof lemma cardinality of indicators (*Discrete_UCB_Step1.thy*)

In the second expression we have:

$$\sum_{t \in \{k \leq t < n\}} \mathbf{1}_{\pi(t, \omega) = a \wedge I(t)} = \text{card}\{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge I(t)\}.$$

Recall the definition of indicator functions:

$$\mathbf{1}_A(t) = \begin{cases} 1 & t \in A \\ 0 & t \notin A \end{cases}.$$

In Isabelle, the indicator function on a predicate set is defined as

$$\text{indicator } A(t) = \text{of_bool}(t \in A).$$

Using this, we rewrite the sum:

Since

$$\text{of_bool}(P) = \begin{cases} 1 & \text{if } P \\ 0 & \text{otherwise} \end{cases}.$$

we rewrite again:

We rewrite the set $\{t \mid k \leq t < n\}$ in terms of Isabelle's interval notation:

$$\{t \mid k \leq t < n\} = \{k, k+1, \dots, n-1\} = \text{atLeastLessThan } k, n.$$

Formally:

Finally, notation $\sum_{t=k}^{n-1} f(t)$ corresponds to

Now we combine all the equalities to conclude the lemma using the **apply** functionality in Isabelle. This lemma provides the fundamental bridge between counting elements of a set defined by a predicate and summing up indicator functions over a finite interval. This equivalence is crucial for analyzing algorithms like UCB where rewards or actions are tallied over time intervals, enabling us to reason about sums rather than directly about set cardinalities.

4.4 A Preliminary Expression on N_n

In this section we consider the expression of N_n that we define in Isabelle as follows:

```

lemma Nn_expression:
  fixes a :: 'a and s :: "nat  $\Rightarrow$  real"
    and k :: nat and n :: nat
  assumes "a  $\in$  A"
    and "k  $\leq$  n"
    and "0 < n"
    and " $\forall t \in \{0..n\}. 0 < s\ t$ "
    and " $\forall t < n - 1. s\ t \leq s\ (t + 1)$ " (* s non-decreasing *)
    and init_play_once: " $\forall \omega. a \in A \longrightarrow N\_n\ k\ a\ \omega = 1$ "
    and finite_played_sets:
      "finite {t. t < n  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
      "finite {t. t < k  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
      "finite {t. k  $\leq$  t  $\wedge$  t < n  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
  shows
    "(N_n n a  $\omega$ ) = 1 + ( $\sum t = k..<n. \text{if } \pi\ (t+1)\ \omega = a \wedge \text{real } (N\_n\ t\ a\ \omega) < s\ t \text{ then } 1 \text{ else } 0$ ) +
      ( $\sum t = k..<n. \text{if } \pi\ (t+1)\ \omega = a \wedge \text{real } (N\_n\ t\ a\ \omega) \geq s\ t \text{ then } 1 \text{ else } 0$ )"
proof - [98 lines]
qed

```

Figure 10: Expression for N_n (*Discrete_UCB_Step1.thy*)

Let $a \in A$, $k, n \in \mathbb{N}$ with $k \leq n$ and $n > 0$. Let $s : \mathbb{N} \rightarrow \mathbb{R}$ be a non-decreasing sequence, i.e.,

$$\forall t < n - 1, \quad s(t) \leq s(t + 1).$$

Suppose further that

$$\forall \omega, \quad N_n(k, a, \omega) = 1,$$

and the sets

$$\{t \mid t < n \wedge \pi(t, \omega) = a\}, \quad \{t \mid t < k \wedge \pi(t, \omega) = a\}, \quad \{t \mid k \leq t < n \wedge \pi(t, \omega) = a\},$$

are all finite.

Recall the definition of N_n :

$$N_n(n, a, \omega) = \text{card}\{t \mid t < n \wedge \pi(t, \omega) = a\}.$$

Next, show the initial count at time k is at least 1, actually equal to 1 by assumption:

$$1 \leq \text{card}\{t \mid t < k \wedge \pi(t, \omega) = a\}.$$

Observe that the set of times less than n where $\pi = a$ splits into two disjoint subsets:

$$\{t < n \mid \pi(t, \omega) = a\} = \{t < k \mid \pi(t, \omega) = a\} \cup \{k \leq t < n \mid \pi(t, \omega) = a\},$$

with disjointness ensured by the inequality ranges.

Since both sets are finite, the cardinality of the union is the sum of cardinalities:

$$\text{card}\{t < n \mid \pi(t, \omega) = a\} = \text{card}\{t < k \mid \pi(t, \omega) = a\} + \text{card}\{k \leq t < n \mid \pi(t, \omega) = a\}.$$

Replace $\text{card}\{t < k \mid \pi(t, \omega) = a\} = N_n(k, a, \omega) = 1$ from the assumption:

$$\text{card}\{t < n \mid \pi(t, \omega) = a\} = 1 + \text{card}\{k \leq t < n \mid \pi(t, \omega) = a\}.$$

```

have eq_card_k: "card {t. t < k ∧ π t ω = a} = N_n k a ω"
  by (simp add: N_n)

have card_k_eq_1: "card {t. t < k ∧ π t ω = a} = 1"
  using init_play_once assms eq_card_k by simp

have card_eq_1:
  "card {t. t < n ∧ π t ω = a} = 1 + card {t. k ≤ t ∧ t < n ∧ π t ω = a}"
  using card_eq card_k_eq_1 eq_card_k by simp

```

Figure 11: Part (i) of proof for N_n expression (*Discrete_UCB_Step1.thy*)

Partition the latter set based on whether $N_n(t, a, \omega) < s(t)$ or $N_n(t, a, \omega) \geq s(t)$:

$$\begin{aligned} \{k \leq t < n \mid \pi(t, \omega) = a\} &= \{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge N_n(t, a, \omega) < s(t)\} \\ &\cup \{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge N_n(t, a, \omega) \geq s(t)\}. \end{aligned}$$

These two subsets are disjoint, so the cardinalities sum.

```

have finite_lt: "finite {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) < s t}"
using finite2 finite_subset by simp

have finite_ge: "finite {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) ≥ s t}"
using finite2 finite_subset by simp

have card_eq_partition:
  "card {t. k ≤ t ∧ t < n ∧ π t ω = a} =
    card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) < s t} +
    card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) ≥ s t}"
proof -
  have union_eq_partition: [3 lines]

  have disj_partition: [3 lines]

  show ?thesis [1 lines]
qed

```

Figure 12: Part (ii) of proof for N_n expression (*Discrete_UCB_Step1.thy*)

Applying the `cardinality_indic_eq` lemma, cardinalities of these sets equal sums over indicators:

$$\text{card}\{t \mid k \leq t < n \wedge \pi(t, \omega) = a \wedge N_n(t, a, \omega) < s(t)\} = \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega)=a \wedge N_n(t, a, \omega) < s(t)},$$

and similarly for the other part.

```

have card_eq_sum_lt:
  "card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) < s t} =
    (∑ t = k..<n. if π t ω = a ∧ real (N_n t a ω) < s t then 1 else 0)"
using cardinality_indic_eq by simp

have card_eq_sum_ge:
  "card {t. k ≤ t ∧ t < n ∧ π t ω = a ∧ real (N_n t a ω) ≥ s t} =
    (∑ t = k..<n. if π t ω = a ∧ real (N_n t a ω) ≥ s t then 1 else 0)"
using cardinality_indic_eq by simp

```

Figure 13: Part (iii) of proof for N_n expression (*Discrete_UCB_Step1.thy*)

The equivalence $x \geq y \iff y \leq x$ (proved by `ge_rewrite`) allows rewriting the second sum's predicate. Combining all steps yields the final equality:

$$N_n(n, a, \omega) = 1 + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega)=a \wedge N_n(t, a, \omega) < s(t)} + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega)=a \wedge N_n(t, a, \omega) \geq s(t)}.$$

```

ultimately have final_eq: "(N_n n a ω) = 1 + (∑ t = k..<n. if π t ω = a ∧ real (N_n t a ω) < s t then 1 else 0) +
then 1 else 0) +
      (∑ t = k..<n. if π t ω = a ∧ real (N_n t a ω) ≥ s t then 1 else 0) "
by simp
show "(N_n n a ω) = 1 + (∑ t = k..<n. if π t ω = a ∧ real (N_n t a ω) < s t then 1 else 0) + [5 lines]
qed

```

Figure 14: Part (iv) of proof for N_n expression (*Discrete_UCB_Step1.thy*)

This proof carefully decomposes the cardinality of the set $\{t < n : \pi(t, \omega) = a\}$ into an initial fixed part and a partition of the later times, using set theory, finiteness, indicator functions, and elementary inequalities. The final expression reflects how N_n can be counted by separating times when the count $N_n(t, a, \omega)$ is below or above a threshold $s(t)$.

4.5 Contradiction Step

In this section we further expand our analysis. In particular, we want to show

$$1 + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega)=a \wedge N_n(t, a, \omega) < s(t)} \leq s(n).$$

Let us develop such a lemma in Isabelle and define it as follows:

```

lemma upper_bound_expression_contradiction:
  fixes a :: 'a and s :: "nat ⇒ real"
  and k :: nat and n :: nat
  and s_n_nat :: nat
  assumes "a ∈ A"
  and "k ≤ n"
  and "0 < n"
  and non_neg_s: "∀ t ∈ {0..n}. 0 < s t"
  and base_le: "s 0 ≤ s 1"
  and non_dec: "∀ t < n - 1. s t ≤ s (t + 1)" (* s is non-decreasing *)
  and s_mono: "∧ t. k ≤ t ∧ t ≤ n ⇒ s t ≤ s n"
  and init_play_once: "∀ ω. a ∈ A ⇒ N_n k a ω = 1"
  and finite_played_sets:
  | "finite {t. t < n ∧ π (t+1) ω = a}"
    "finite {t. t < k ∧ π (t+1) ω = a}"
    "finite {t. k ≤ t ∧ t < n ∧ π (t+1) ω = a}"
  and xs_sorted_def: "xs = sorted_list_of_set {t ∈ {k..<n}. π (t+1) ω = a ∧ real (N_n t a ω) < s t}"
  and s_nat_def: "s_n_nat = nat (⌊s n⌋)"
  and len_bound_def: "s_n_nat < length xs" (* s_n_nat is an existing element *)
  and distinct_xs: "distinct xs"
  and gt_ineq: "length xs + 1 > ⌊s n⌋"
  and N_n_increasing_with_plays:
  "∀ t t'. k ≤ t ∧ t < t' ∧ π (t+1) ω = a ∧ π (t'+1) ω = a ⇒ N_n t' a ω ≥ N_n t a ω + 1"
  and neg: "1 + real (∑ t=k..<n. if π (t+1) ω = a ∧ real (N_n t a ω) < s t then 1 else 0) > s n"
  and "t_hat ∈ set xs"
  and "t_hat = xs ! s_n_nat"
  and "real (N_n t_hat a ω) ≥ real s_n_nat + 1"
  (* By construction of xs, and since t_hat is the s_n_nat-th element in xs,
    the count N_n t_hat a ω is at least s_n_nat + 1 *)
  (* The argument depends on repeated increments due to monotonicity *)
  shows "(real (N_n t_hat a ω) ≥ ⌊s n⌋ + 1) ∧ (π (t_hat+1) ω = a ∧ (real (N_n t_hat a ω) < s t_hat))"

```

Figure 15: Proof by contradiction problem statement (*Discrete_UCB_Step1.thy*)

Let $a \in A$, meaning that the action a belongs to the set of all possible actions A . The index k is assumed to satisfy $k \leq n$, and we further assume that $n > 0$, so the time horizon is strictly positive. We are given a sequence $s(t)$ defined over the discrete time domain $\{0, \dots, n\}$. We assume that $s(t)$ is strictly positive over this domain:

$$\forall t \in \{0, \dots, n\}, \quad 0 < s(t).$$

Moreover, the sequence starts in a non-decreasing fashion:

$$s(0) \leq s(1),$$

and continues to be non-decreasing over the interval:

$$\forall t < n - 1, \quad s(t) \leq s(t + 1).$$

The values of $s(t)$ on the interval $[k, n]$ are assumed to be bounded above by $s(n)$:

$$k \leq t \leq n \Rightarrow s(t) \leq s(n).$$

We suppose that if an action a belongs to A , then it is played exactly once at time k , formally:

$$a \in A \Rightarrow N_n(k, a, \omega) = 1.$$

Additionally, we impose conditions on the number of times an action a occurs in various time intervals. Specifically:

$$\text{finite } \{t < n \mid \pi(t, \omega) = a\}, \quad \text{finite } \{t < k \mid \pi(t, \omega) = a\}, \quad \text{finite } \{k \leq t < n \mid \pi(t, \omega) = a\},$$

implying that the occurrences of action a are finite in each of these intervals.

Let xs be defined as the `sorted_list_of_set` of times when the action a was played and the associated play count was under some threshold. This list is assumed to be distinct:

$$\text{distinct}(xs),$$

and of a length sufficient to accommodate the natural floor of $s(n)$:

$$s_{\text{n_nat}} = \lfloor s(n) \rfloor, \quad s_{\text{n_nat}} < \text{length}(xs), \quad \text{length}(xs) + 1 > \lfloor s(n) \rfloor.$$

We assume that:

$$\forall t < t', \quad \text{the play count increases by at least one between repeated plays of } a,$$

and that the total number of plays where the play count remains below the threshold is too large:

$$1 + \sum \dots > s(n).$$

Let $\hat{t}$ be the time chosen from the list xs corresponding to the index $\lfloor s(n) \rfloor$:

$$\hat{t} \in \text{set}(xs), \quad \hat{t} = xs[s_{\text{n_nat}}].$$

By construction, we have:

$$\hat{t} \in xs \subseteq \{t \in \{k, \dots, n-1\} \mid \pi(t, \omega) = a \wedge N_n(t, a, \omega) < s(t)\}.$$

Hence,

$$\pi(\hat{t}, \omega) = a \quad \text{and} \quad N_n(\hat{t}, a, \omega) < s(\hat{t}).$$

This is shown in the below section of the proof

```
have part_1_glob: " $\pi \ t\_hat \ \omega = a \wedge \text{real } (N\_n \ t\_hat \ a \ \omega) < s \ t\_hat$ "
proof - [36 lines]
qed
```

Figure 16: Proof by contradiction key result (i) (*Discrete_UCB_Step1.thy*)

However, due to the definition of xs and the assumption that xs has at least $s_n_nat + 1$ elements, and from the monotonicity of N_n as new a -plays occur (each increases N_n by at least 1), the count $N_n(\hat{t}, a, \omega)$ must be at least $s_n_nat + 1$:

$$N_n(\hat{t}, a, \omega) \geq s_n_nat + 1 = \lfloor s(n) \rfloor + 1.$$

```
have final_result: " $\text{real } (N\_n \ t\_hat \ a \ \omega) \geq \lfloor s \ n \rfloor + 1$ " [2 lines]
```

```
show ?thesis using final_result and part_1_glob by auto
qed
```

Figure 17: Proof by contradiction key result (ii) (*Discrete_UCB_Step1.thy*)

This contradicts the earlier fact that:

$$N_n(\hat{t}, a, \omega) < s(\hat{t}) \leq s(n).$$

Thus,

$$N_n(\hat{t}, a, \omega) < s(n) \quad \text{and} \quad N_n(\hat{t}, a, \omega) \geq \lfloor s(n) \rfloor + 1,$$

cannot both hold.

Therefore, a contradiction arises. Hence, the assumption that there are more than $\lfloor s(n) \rfloor$ plays of action a between times k and $n - 1$ with count less than the threshold $s(t)$ must be false.

This completes the proof by contradiction, establishing the upper bound:

$$1 + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t, \omega) = a \wedge N_n(t, a, \omega) < s(t)} \leq s(n).$$

4.6 A Bound on N_n

Combining all previous results and we assume that $1 + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t,\omega)=a \wedge N_n(t,a,\omega) < s(t)} \leq s(n)$ holds based on our proof by contradiction we implement the following lemma:

```

lemma Nn_upper_bound:
  fixes a :: 'a and s :: "nat  $\Rightarrow$  real"
  and k :: nat and n :: nat
  assumes asm: "real(1 + ( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  real (N_n t a  $\omega$ ) < s t then 1 else 0))  $\leq$  s n"
  and a_in_A: "a  $\in$  A"
  and k_le_n: "k  $\leq$  n"
  and n_pos: "0 < n"
  and s_pos: " $\forall$  t  $\in$  {0.. $n$ }. 0 < s t"
  and s_nondec: " $\forall$  t < n - 1. s t  $\leq$  s (t + 1)"
  and init_play_once: " $\forall \omega$ . a  $\in$  A  $\rightarrow$  N_n k a  $\omega$  = 1"
  and finite_played_sets_1: "finite {t. t < n  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
  and finite_played_sets_2: "finite {t. t < k  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
  and finite_played_sets_3: "finite {t. k  $\leq$  t  $\wedge$  t < n  $\wedge$   $\pi$  (t+1)  $\omega$  = a}"
  shows "real(N_n n a  $\omega$ )  $\leq$  s n + real(( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$  real (N_n t a  $\omega$ ) then 1 else 0))"
proof - [20 lines]
qed

```

Figure 18: N_n upper bound expression (*Discrete_UCB_Step1.thy*)

Hence, we have verified that:

$$N_n(n, a, \omega) \leq s_n + \sum_{t=k}^{n-1} \mathbf{1}_{\pi(t,\omega)=a \wedge N_n(t,a,\omega) \geq s(t)}.$$

For taking expectations at both sides we need to define the following lemma:

```

theorem ENn_upper_bound:
  assumes
    a_in_A: "a  $\in$  A"
    and k_le_n: "k  $\leq$  n"
    and n_pos: "0 < n"
    and s_pos: " $\forall$  t  $\in$  {0.. $n$ }. 0 < s t"
    and s_nondec: " $\forall$  t < n. s t  $\leq$  s (t + 1)"
    and init_play_once: " $\forall \omega$ . a  $\in$  A  $\rightarrow$  N_n k a  $\omega$  = 1"
    and integrable_Nn: "integrable M ( $\lambda \omega$ . real (N_n n a  $\omega$ ))"
    and integrable_rhs_sum: "integrable M ( $\lambda \omega$ . s n + ( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$ 
     $\leq$  real (N_n t a  $\omega$ ) then 1 else 0))"
    and integrable_s: "integrable M ( $\lambda \omega$ . s n)"
    and integrable_indicator_sum: "integrable M ( $\lambda \omega$ .  $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$ 
    real (N_n t a  $\omega$ ) then 1 else 0)"
    and linearity: "integralL M ( $\lambda \omega$ . s n + ( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$  real (N_n t a  $\omega$ )
    then 1 else 0)) =
    integralL M ( $\lambda \omega$ . s n) + integralL M ( $\lambda \omega$ .  $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$ 
    real (N_n t a  $\omega$ ) then 1 else 0)"
    and pointwise_bound: "real (N_n n a  $\omega$ )  $\leq$  s n + ( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$  real
    (N_n t a  $\omega$ ) then 1 else 0)"
    and mono_intgrl: "integralL M ( $\lambda \omega$ . real (N_n n a  $\omega$ ))  $\leq$  integralL M ( $\lambda \omega$ . s n + ( $\sum$ t = k.. $n$ .
    if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$  real (N_n t a  $\omega$ ) then 1 else 0))"
  shows
    "expectation ( $\lambda \omega$ . real (N_n n a  $\omega$ ))  $\leq$ 
    s n + expectation ( $\lambda \omega$ . ( $\sum$ t = k.. $n$ . if  $\pi$  (t+1)  $\omega$  = a  $\wedge$  s t  $\leq$  real (N_n t a  $\omega$ ) then 1 else 0))"
proof - [54 lines]

```

Figure 19: Expectation of N_n upper bound (*Discrete_UCB_Step1.thy*)

In order to complete the proof of this lemma we need to define and combine several assumptions such as integrability of the LHS and RHS, monotonicity and linearity of integrals. The complete proof is stated below:

```

proof -
  let ?f = "λω. real (N n n a ω)"
  let ?g = "λω. s n + (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N n t a ω) then 1 else 0)"
  let ?g1 = "λω. s n"
  let ?g2 = "λω. (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N n t a ω) then 1 else 0)"
  have pointwise_le: "?f ω ≤ ?g ω" [1 lines]

  have intg_f: "integrable M ?f" using integrable_Nn .
  have intg_g: "integrable M ?g" using integrable_rhs_sum .
  then have eq_f: "integral^L M ?f = expectation (?f)" [1 lines]
  then have eq_g: "integral^L M ?g = expectation (?g)" [1 lines]

  then have sub_eqg1: "integral^L M ?g1 = expectation (λω. s n)" [1 lines]
  then have sub_eqg2: "integral^L M ?g2 = expectation (λω. (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N n t a ω) then 1 else 0))" [1 lines]
  by simp

  have "real (N n n a ω) ≤ s n + (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N n t a ω) then 1 else 0)" [1 lines]

  have const_measure: "Sigma_Algebra.measure M (space M) = 1" [1 lines]

  have exp_sn: "expectation (λω. s n) = s n" [7 lines]

  have rhs_expr: [2 lines]

  then have rhs_expr: [2 lines]

  also have final_eq: "expectation (?g) = s n + expectation (λω. (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N n t a ω) then 1 else 0))" [1 lines]
  using rhs_expr exp_sn
  by simp

  have glob_final_eq: "expectation (?f) ≤ expectation (?g)" [2 lines]

  thus ?thesis [1 lines]

```

Figure 20: Proof expectation of N_n upper bound (*Discrete_UCB_Step1.thy*)

This lemma verifies:

$$\mathbb{E}[N_{n,a}] \leq s_n + \mathbb{E}\left[\sum_{t=k}^{n-1} \mathbf{1}_{\pi(t,\omega)=a \wedge N_n(t,a,\omega) \geq s(t)}\right].$$

It then follows by definition of the expectation of indicator functions that:

$$\mathbb{E}\left[\sum_{t=k}^{n-1} \mathbf{1}_{\pi(t,\omega)=a \wedge N_n(t,a,\omega) \geq s(t)}\right] = \sum_{t=k}^{n-1} \mathbb{P}(A_{t+1} = a, N_{t,a} \geq s_t).$$

By the definition of conditional probabilities we get: $\sum_{t=k}^{n-1} \mathbb{P}(A_{t+1} = a, N_{t,a} \geq s_t) = \sum_{t=k}^{n-1} \mathbb{P}(A_{t+1} = a | N_{t,a} \geq s_t) \mathbb{P}(N_{t,a} \geq s_t)$. Hence, we have completed the verification of the expression:

$$\mathbb{E}[N_{n,a}] \leq s_n + \sum_{t=s_k}^{n-1} \mathbb{P}(A_{t+1} = a | N_{t,a} \geq s_t).$$

This tells us that for any Bandit algorithm $\mathbb{E}[N_{n,a}]$ is bounded by the sum of the probabilities that arm a is played given that it has been played for a certain number of times in the past.

5 Step 2: Discrete UCB algorithm Proof Verification

This section guides us through the second step of the discrete UCB proof of Rebeschini (2022) using our definitions and results from proposition 15.1 and our analysis in previous sections regarding obtaining an upper bound for $\mathbb{E}[N_{n,a}]$.

We first define several key statements for our upper-bound definition. The sample mean M estimates the average reward for an arm based on observed outcomes. It is defined as the average of N_n i.i.d. reward samples collected up to time t :

$$M(t, a, \omega) = \frac{1}{N_n(t, a, \omega)} \sum_{i < N_n(t, a, \omega)} Z(i, a, \omega) \cdot I_{\pi(t, \omega)=a}.$$

The upper confidence bound (UCB) value for arm a at time t , under outcome ω , is defined as:

$$U(t, a, \omega) = M(t, a, \omega) + q \cdot \sqrt{\frac{\log(1/\delta)}{2 \cdot N_n(t, a, \omega)}},$$

where q is a confidence parameter and N_n denotes the number of times arm a has been selected by time t .

The arm selected at time $t + 1$ is the one with the maximum UCB value:

$$A_{t+1}(\omega) = \arg \max_{a \in A} U(t, a, \omega).$$

This is implemented in Isabelle/HOL using the **SOME** operator, which selects a maximum element from a finite set:

$$A_{t+1}(\omega) = \text{SOME } a \in A. \forall a' \in A. U(t, a', \omega) \leq U(t, a, \omega).$$

Let $\mathcal{P}$ be a finite probability measure on space Ω . Then, for any measurable sets $A \subseteq B$, the following inequality holds:

$$\text{prob}_{\mathcal{P}}(A) \leq \text{prob}_{\mathcal{P}}(B).$$

This expresses the monotonicity of finite probability measures and is used to bound probabilities of subsets in the analysis. In Isabelle this is defined shown in figure below:

```

locale bandit_policy = bandit_problem + prob_space +
  fixes  $\Omega$  :: "'b set"
  and  $\mathcal{F}$  :: "'b set set"
  and  $\omega$  :: 'b
  fixes  $\pi$  :: "nat  $\Rightarrow$  'b  $\Rightarrow$  'a"
and  $N_n$  :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  nat"
and  $Z$  :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  real"
and  $\delta$  :: real
and  $q$  :: real
assumes finite_A: "finite A"
  and a_in_A: "a  $\in$  A"
  and measurable_policy: " $\forall t. \pi t \in$  measurable P (count_space A)"
  and  $N_n$  def: " $\forall n a b. N_n n a b = \text{card } \{t \in \{0..<n\}. \pi (t+1) b = a\}$ "
  and  $\delta_{\text{pos}}$ : " $0 < \delta$ "
  and  $\delta_{\text{less1}}$ : " $\delta < 1$ "
  and  $q_{\text{pos}}$ : " $q > 0$ "
begin
definition sample_mean_Z :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  real" where
  "sample_mean_Z n a b  $\equiv$  (1 / real n) * ( $\sum i < n. Z i a b$ )"

definition M_val :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  real" where
  "M_val t a b  $\equiv$  (if  $N_n (t+1) a b = 0$  then 0
    else ( $\sum s < t. \text{if } \pi s b = a \text{ then } Z s a b \text{ else } 0$ ) / real ( $N_n t a b$ ))"

definition U :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  real" where
  "U t a b  $\equiv$  M_val t a b +  $q * \text{sqrt} (\ln (1 / \delta) / (2 * \text{real} (\max 1 (N_n t a b))))$ "

definition A_t_plus_1 :: "nat  $\Rightarrow$  'b  $\Rightarrow$  'a" where
  "A_t_plus_1 t b  $\equiv$  (SOME a. a  $\in$  A  $\wedge$  ( $\forall a'. a' \in A \longrightarrow U t a b \geq U t a' b$ ))"

lemma (in finite_measure) finite_measure_mono:
  assumes "A  $\subseteq$  B" "B  $\in$  sets M" shows "measure M A  $\leq$  measure M B"
  using emeasure_mono[OF assms] emeasure_real[of A] emeasure_real[of B] bv (auto simp: measure_def)

```

Figure 21: Key definitions and a monotonicity lemma of probability measures (*Discrete_UCB_Step2.thy*)

5.1 Union Bound

The union bound is a key result in probability theory that we require for verifying the following expression as part of the second step of the discrete UCB proof verification

$$\mathbb{P}(A_{t+1} = a \mid N_{t,a} \geq s) \leq \mathbb{P}(U_{t,a^*} \leq \mu_{a^*} \mid N_{t,a} \geq s) + \mathbb{P}(\mu_{a^*} \leq U_{t,a} \mid N_{t,a} \geq s).$$

This result will be further discussed in section 5.5. For now, the union bound verification is implemented as the following theorem in Isabelle:

```

theorem union_bound:
  fixes E F G :: "'b set"
  assumes "E ⊆ F ∪ G"
  and "E ∈ events" "F ∈ events" "G ∈ events"
  shows "prob E ≤ prob F + prob G"
proof -
  have "F ∪ G ∈ events"
  using assms(3,4) sets.Un by blast
  have "prob E ≤ prob (F ∪ G)"
  using assms local.finite_measure_mono by auto
  also have "prob (F ∪ G) ≤ prob F + prob G"
  using assms local.finite_measure_subadditive by blast
  finally show ?thesis .
qed

```

Figure 22: Proof union bound (*Discrete_UCB_Step2.thy*)

For this proof we use general definitions and properties of probability theory from Isabelle's prob-space.

5.2 Hoeffding i.i.d. Bounds for General Bounded Random Variables

Let $a \in A$, $n \in \mathbb{N}$, $\varepsilon \geq 0$, $\hat{\mu} \in \mathbb{R}$, and real numbers $l < u$ be given. Consider the sequence of random variables $Z_i(a) : \Omega \rightarrow \mathbb{R}$ for $i < n$ defined on a probability space $(\Omega, \mathcal{F}, M)$, with the following properties: each $Z_i(a)$ is almost surely bounded between l and u , i.e.,

$$l \leq Z_i(a)(\omega) \leq u \quad \text{for all } \omega \in \Omega,$$

the variables are independent, and

$$\hat{\mu} = \sum_{i < n} \mathbb{E}[Z_i(a)].$$

We aim to show that the probability of the sum exceeding its expectation by ε satisfies

$$\mathbb{P}\left(\sum_{i < n} Z_i(a) \geq \hat{\mu} + \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right).$$

Define the index set $I = \{i \mid i < n\}$ and the functions $X_i = Z_i(a)$, as well as the bounds $a_i = l$ and $b_i = u$. The index set I is finite, which allows us to apply Hoeffding's inequality. For each $i \in I$, the boundedness assumption implies

$$a_i \leq X_i(\omega) \leq b_i \quad \text{almost everywhere in } M,$$

ensuring the almost-everywhere boundedness condition required for Hoeffding's inequality.

Since the variables $Z_i(a)$ are independent and each $Z_i(a)$ is a measurable random variable with respect to the Borel σ -algebra, we may invoke the Hoeffding inequality for independent bounded random variables. Specifically, define the sum of squared widths

$$\sum_{i \in I} (b_i - a_i)^2 = \sum_{i < n} (u - l)^2 = n(u - l)^2,$$

which is strictly positive because $u \neq l$ and $n > 0$.

Next, observe that the sum of expectations can be expressed as the sum of Lebesgue integrals:

$$\sum_{i < n} \mathbb{E}[Z_i(a)] = \sum_{i \in I} \int_{\Omega} X_i dM = \hat{\mu}.$$

Applying the one-sided Hoeffding inequality, we obtain

$$\mathbb{P}\left(\sum_{i \in I} X_i - \sum_{i \in I} \int_{\Omega} X_i dM \geq \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i \in I} (b_i - a_i)^2}\right).$$

Substituting the previous expressions for the sum of expectations and the sum of squared widths, we rewrite the inequality in the original variables as

$$\mathbb{P}\left(\sum_{i < n} Z_i(a) \geq \hat{\mu} + \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{n(u - l)^2}\right),$$

which completes the proof. In Isabelle we implement this as follows:

```

theorem hoeffding_iid_bound_ge_general:
  fixes a :: 'a and n :: nat and ε :: real and μ_hat :: real and l u :: real
  assumes a_in: "a ∈ A"
    and eps_pos: "ε ≥ 0"
    and bounds: "∀i < n. ∀ω ∈ Ω. l ≤ Z i a ω ∧ Z i a ω ≤ u"
    and mu_def: "μ_hat = (∑ i < n. expectation (λω. Z i a ω))"
    and "u - l ≠ 0"
    and n_pos: "n > 0"
    and space_M: "space M = Ω"
    and sets_M: "sets M = ℱ"
    and indep: "indep_vars (λ_. borel) (λi. (λω. Z i a ω)) {i. i < n}"
    and rv: "∀i < n. random_variable borel (λω. Z i a ω)"

  shows "prob {ω ∈ Ω. (∑ i < n. Z i a ω) ≥ μ_hat + ε}
    ≤ exp (- 2 * ε^2 / (real n * (u - l)^2))"
proof - [62 lines]
qed

```

Figure 23: A general Hoeffding bound(i) (*Discrete_UCB_Step2.thy*)

Let $a \in A$, $n \in \mathbb{N}$, $\varepsilon \geq 0$, $\hat{\mu} \in \mathbb{R}$, and real numbers $l < u$ be given. Consider a sequence of random variables $Z_i(a) : \Omega \rightarrow \mathbb{R}$ for $i < n$ defined on a probability space $(\Omega, \mathcal{F}, M)$,

satisfying (i) Almost sure bounds: $l \leq Z_i(a)(\omega) \leq u$ for all $\omega \in \Omega$, (ii) the $Z_i(a)$ are independent, (iii) $\hat{\mu} = \sum_{i < n} \mathbb{E}[Z_i(a)]$.

We aim to bound the probability that the sum deviates below its expectation by at least ε :

$$\mathbb{P}\left(\sum_{i < n} Z_i(a) \leq \hat{\mu} - \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right).$$

Define the index set $I = \{i \mid i < n\}$, the functions $X_i = Z_i(a)$, and the bounds $a_i = l$, $b_i = u$. The finiteness of I allows us to apply Hoeffding's inequality. For each $i \in I$, the almost-everywhere boundedness condition

$$a_i \leq X_i(\omega) \leq b_i \quad \text{for almost every } \omega \in M,$$

is satisfied by the original bounds on $Z_i(a)$.

Because the $Z_i(a)$ are independent and measurable, we may invoke Hoeffding's inequality for independent bounded random variables. The sum of squared widths is

$$\sum_{i \in I} (b_i - a_i)^2 = \sum_{i < n} (u - l)^2 = n(u - l)^2 > 0.$$

The sum of expectations can be expressed as the sum of Lebesgue integrals:

$$\sum_{i < n} \mathbb{E}[Z_i(a)] = \sum_{i \in I} \int_{\Omega} X_i dM = \hat{\mu}.$$

Applying the lower-tail version of Hoeffding's inequality, we obtain

$$\mathbb{P}\left(\sum_{i \in I} X_i - \sum_{i \in I} \int_{\Omega} X_i dM \leq -\varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i \in I} (b_i - a_i)^2}\right).$$

Substituting the sums of expectations and squared widths back to the original variables gives

$$\mathbb{P}\left(\sum_{i < n} Z_i(a) \leq \hat{\mu} - \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right),$$

which completes the proof. In Isabelle this theorem is defined as follows:

```

theorem hoeffding_iid_bound_le_general:
  fixes a :: 'a and n :: nat and ε :: real and μ_hat :: real and l u :: real
  assumes a_in: "a ∈ A"
    and eps_pos: "ε ≥ 0"
    and bounds: "∀i < n. ∀ω ∈ Ω. l ≤ Z i a ω ∧ Z i a ω ≤ u"
    and mu_def: "μ_hat = (∑ i < n. expectation (λω. Z i a ω))"
    and "u - l ≠ 0"
    and n_pos: "n > 0"
    and space_M: "space M = Ω"
    and sets_M: "sets M = ℱ"
    and indep: "indep_vars (λ_. borel) (λi. (λω. Z i a ω)) {i. i < n}"
    and rv: "∀i < n. random_variable borel (λω. Z i a ω)"
  shows "prob {ω ∈ Ω. (∑ i < n. Z i a ω) ≤ μ_hat - ε}
    ≤ exp (- 2 * ε^2 / (real n * (u - l)^2))"
proof - [73 lines]
qed

```

Figure 24: A general Hoeffding bound(ii) (*Discrete_UCB_Step2.thy*)

5.3 Hoeffding's Inequality: A Concentration Bound for Sum of i.i.d Random Variables

5.3.1 Hoeffding Lower-Tail Bound

Let $\{\hat{Z}_{j,a^*}\}_{j=0}^{n-1}$ be independent real-valued random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Assume there exist known constants l and u with $l \leq \hat{Z}_{j,a^*}(\omega) \leq u$ for all $j < n$ and a.e. ω . Let

$$\hat{\mu} := \sum_{j=0}^{n-1} \mathbb{E}[\hat{Z}_{j,a^*}].$$

Fix $\delta \in (0, 1]$ and define

$$\varepsilon := |u - l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}.$$

Then

$$\mathbb{P}\left\{\sum_{j=0}^{n-1} \hat{Z}_{j,a^*} \leq \hat{\mu} - \varepsilon\right\} \leq \delta.$$

For independent random variables X_j with almost sure bounds $a_j \leq X_j \leq b_j$, Hoeffding's inequality (lower-tail version) states that for any $\varepsilon \geq 0$,

$$\mathbb{P}\left\{\sum_{j=0}^{n-1} X_j \leq \sum_{j=0}^{n-1} \mathbb{E}[X_j] - \varepsilon\right\} \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{j=0}^{n-1} (b_j - a_j)^2}\right). \quad (\text{H})$$

Take $X_j := \hat{Z}_{j,a^*}$, and $a_j := l$, $b_j := u$ for all j . By the assumption $l \leq \hat{Z}_{j,a^*} \leq u$ a.s. and independence, (H) yields for any $\varepsilon \geq 0$

$$\mathbb{P}\left\{\sum_{j=0}^{n-1} \hat{Z}_{j,a^*} \leq \hat{\mu} - \varepsilon\right\} \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{j=0}^{n-1} (u - l)^2}\right). \quad (1)$$

Since the bounds are identical for each j ,

$$\sum_{j=0}^{n-1} (u - l)^2 = n(u - l)^2. \quad (2)$$

Substituting (2) into (1) gives

$$\mathbb{P}\left\{\sum_{j=0}^{n-1} \hat{Z}_{j,a^*} \leq \hat{\mu} - \varepsilon\right\} \leq \exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right). \quad (3)$$

Set $\varepsilon = |u - l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}$. Then

$$\varepsilon^2 = (u - l)^2 \cdot \frac{n}{2} \ln \frac{1}{\delta}. \quad (4)$$

Substitute (4) into the exponent of (3):

$$-\frac{2\varepsilon^2}{n(u-l)^2} = -\frac{2(u-l)^2 \cdot \frac{n}{2} \ln \frac{1}{\delta}}{n(u-l)^2} = -\ln \frac{1}{\delta}. \quad (5)$$

From (3) and (5),

$$\mathbb{P}\left\{\sum_{j=0}^{n-1} \hat{Z}_{j,a^*} \leq \hat{\mu} - \varepsilon\right\} \leq \exp(-\ln(1/\delta)) = \delta,$$

which proves the claim. The complete proof is given in Isabelle as stated below:

```

theorem hoeffding_iid_bound_le:
  fixes a :: 'a and n :: nat and ε :: real and μ_hat :: real and l_hat u_hat :: real
  and I :: "nat set"
  and X_new :: "nat ⇒ 'b ⇒ real"
  and a_bound b_bound :: "nat ⇒ real"
  assumes a_in: "a ∈ A"
  and "b ∈ Ω" and eps_pos: "ε ≥ 0"
  and eps: "ε = abs (u_hat - l_hat) * sqrt (((real n) / 2) * ln (1 / δ))"
  and "δ ≥ 0 ∧ δ ≤ 1"
  and t_eq_n: "t = n"
  and "c > 0"
  and bounds: "∀ j < t. ∀ ω ∈ Ω. ∀ a ∈ A. l_hat ≤ Z_hat j a ω ∧ Z_hat j a ω ≤ u_hat"
  and mu_def: "μ_hat = (∑ j < t. expectation (λω. Z_hat j a ω))"
  and "u_hat - l_hat ≠ 0"
  and t_pos: "t > 0"
  and "∀ t < n. N n t a_star b > 0"
  and n_pos: "n > 0"
  and "M_val t a_star b ≡ (∑ s < t. if π s b = a_star then Z s a_star b else 0) /
real (N n t a_star b)"
  and widths: "(∑ i ∈ I. (b_bound i - a_bound i)^2) = (real n) * (u_hat - l_hat)^2"
  and space_M: "space M = Ω"
  and sets_M: "sets M = ℱ"
  and indep: "indep_vars (λ_. borel) (λj. (λω. Z j a ω)) {j. j < t}"
  and rv: "∀ j < t. random_variable borel (λω. Z j a ω)"
  and "∀ j < t. Z_hat j a_star ω = c * (if π j b = a_star then Z j a_star b else 0)"
  and "indep_interval_bounded_random_variables M I X_new a_bound b_bound"
  and indep_loc: "indep_interval_bounded_random_variables M {j. j < t} (λj. (λω. Z_hat j a_star ω)) (λj. l_hat)"
  and H: "Hoeffding_ineq M {j. j < t} (λj. (λω. Z_hat j a_star ω)) (λj. l_hat) (λj. u_hat)"
  and sum_integrals_eq: "(∑ j ∈ {j. j < t}. integral^1 M (λω. Z_hat j a_star ω)) = μ_hat"
  and rewriting: "prob {ω ∈ Ω. (∑ j < t. Z_hat j a_star ω) - (∑ j < t. expectation (Z_hat j a_star)) ≤ - ε} =
  prob {x ∈ space M. (∑ j < t. Z_hat j a_star x) ≤ (∑ j < t. expectation (Z_hat j a_star)) - ε}"
  shows "prob {ω ∈ Ω. (∑ j < n. Z_hat j a_star ω) ≤ μ_hat - ε} ≤ δ"
proof - [128 lines]

```

Figure 25: An implementation of Hoeffding bound(i) (*Discrete_UCB_Step2.thy*)

The theorem above is applied to the particular action a^* and to the family of random variables $\{\hat{Z}_{j,a^*}\}_{j=0}^{n-1}$, where in our setup $\hat{Z}_{j,a}(\omega) = c \cdot \mathbf{1}\{A_j(\omega) = a\} Z_{j,a}(\omega)$. The bounds l and u appearing in the statement are the same constants that you assume satisfy $l \leq \hat{Z}_{j,a^*} \leq u$ almost surely for every j . Also, the sum of expectations is

$$\hat{\mu} = \sum_{j=0}^{n-1} \mathbb{E}[\hat{Z}_{j,a^*}],$$

Because each coordinate has the same width $u - l$,

$$\sum_{j=0}^{n-1} (b_j - a_j)^2 = \sum_{j=0}^{n-1} (u - l)^2 = n(u - l)^2,$$

and this is the algebraic step that yields the compact closed-form expression $\exp(-2\varepsilon^2/(n(u-l)^2))$ in Hoeffding's bound.

$\varepsilon = |u - l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}$, one obtains:

$$-\frac{2\varepsilon^2}{n(u-l)^2} = -\ln \frac{1}{\delta} = \delta.$$

Hence, the Hoeffding upper bound equals δ . If we take $c = \frac{1}{N_{t,a}}$ to obtain the empirical mean $M_{t,a} = \frac{1}{N_{t,a}} \sum_{s=1}^t Z_{s,a} \mathbf{1}\{A_s = a\}$, then note that $N_{t,a}$ is *random*; therefore the normalized variables. Also, $\hat{Z}_{j,a} = \frac{1}{N_{t,a}} \mathbf{1}\{A_j = a\} Z_{j,a}$ are only well-defined when $N_{t,a} > 0$. Furthermore, a standard way to proceed is to condition on the event $\{N_{t,a} > 0\}$ (or more generally on $\{N_{t,a} = m\}$) and apply Hoeffding's inequality on that event where the normalization is deterministic. Alternatively, one can work with the unnormalized sum $\sum_{s=1}^t \mathbf{1}\{A_s = a\} Z_{s,a}$, apply Hoeffding there (with bounds scaled by the same factor 1 per occurrence), and then divide the resulting concentration inequality by $N_{t,a}$ on the event $N_{t,a} > 0$.

In further formal verification of the proof one should verify the independence of the family $\{\hat{Z}_{j,a^*}\}_{j < n}$ (or conditional independence when appropriate), that the bounds $l \leq \hat{Z}_{j,a^*} \leq u$ hold almost everywhere (AE) with respect to the probability space, and that each $\hat{Z}_{j,a^*}$ is a measurable (random) variable. If there is a positive probability that $N_{t,a} = 0$, explicitly handle that case by either stating that the inequality conditioned on $\{N_{t,a} > 0\}$ or define the empirical mean only on the event $N_{t,a} > 0$.

Our proof verification already follows the standard structure for applying the Isabelle/Hoeffding lemma and: (i) asserts AE bounds for each coordinate, (ii) shows the independence assumption matches the Hoeffding precondition, (iii) compute $\sum (b_j - a_j)^2 = n(u - l)^2$, (iii) substitute the chosen ε and simplify the exponent to $-\ln(1/\delta)$. The only additional care is the treatment of the random normalization $N_{t,a}$ (if used): either condition on it or replace it by a deterministic value inside the scope of the Hoeffding application.

5.3.2 Hoeffding Upper-Tail Bound

The argument for the upper-tail is symmetric to the lower-tail theorem. The only difference is the sign of the deviation: here we control the probability that the sum exceeds its expectation by more than ε . The cancellation in the exponent is identical: the denominator is $n(u - l)^2$, and with the chosen ε , the bound simplifies to δ . As in the lower-tail case, if one uses the empirical mean normalization $c = 1/N_{t,a}$, one must handle the event $N_{t,a} > 0$ carefully, either by conditioning or by restricting the domain of definition of the mean.

The proof for the upper tail bound is defined in Isabelle as shown below:

```

theorem hoeffding_iid_bound_ge:
  fixes a :: 'a and n :: nat and ε :: real and μ_hat :: real and l_hat u_hat :: real
  and I :: "nat set"
  and X_new :: "nat ⇒ 'b ⇒ real"
  and a_bound b_bound :: "nat ⇒ real"
  assumes a_in: "a ∈ A"
  and "b ∈ Ω" and eps_pos: "ε ≥ 0" and eps: "ε = abs (u_hat - l_hat) * sqrt (((real n) / 2) * ln (1 / δ))"
  and "δ ≥ 0 ∧ δ ≤ 1" and t_eq_n: "t = n" and "c > 0" and bounds: "∀ j < t. ∀ ω ∈ Ω. ∀ a ∈ A. l_hat ≤ Z_hat j a"
  and mu_def: "μ_hat = (∑ j < t. expectation (λ ω. Z_hat j a ω))"
  and "u_hat - l_hat ≠ 0"
  and t_pos: "t > 0"
  and "∀ t < n. N_n t a_star b > 0"
  and n_pos: "n > 0"
  and "M_val t a b ≡ (∑ s < t. if π s b = a then Z s a b else 0) /
  real (N_n t a b)"
  and widths: "(∑ i ∈ I. (b_bound i - a_bound i)^2) = (real n) * (u_hat - l_hat)^2"
  and space M: "space M = Ω"
  and sets_M: "sets M = F"
  and indep: "indep_vars (λ j. (λ ω. Z j a ω)) {j. j < t}"
  and rv: "∀ j < t. random_variable borel (λ ω. Z j a ω)"
  and "∀ j < t. Z_hat j a ω = c * (if π j b = a then Z j a_star b else 0)"
  and "indep_interval_bounded_random_variables M I X_new a_bound b_bound"
  and indep_loc: "indep_interval_bounded_random_variables M {j. j < t}
    (λ j. (λ ω. Z_hat j a ω))
    (λ j. l_hat) (λ j. u_hat)"
  and H: "Hoeffding_ineq M {j. j < t}
    (λ j. (λ ω. Z_hat j a ω))
    (λ j. l_hat) (λ j. u_hat)"
  and sum_integrals_eq: "(∑ j ∈ {j. j < t}. integral^L M (λ ω. Z_hat j a ω)) = μ_hat"
  and rewriting: "prob {ω ∈ Ω. (∑ j < t. Z_hat j a ω) - (∑ j < t. expectation (Z_hat j a)) ≥ ε} =
    prob {x ∈ space M. (∑ j < t. Z_hat j a x) ≥ (∑ j < t. expectation (Z_hat j a)) + ε}"
  shows "prob {ω ∈ Ω. (∑ j < n. Z_hat j a ω) ≥ μ_hat + ε} ≤ δ"
proof - [126 lines]

```

Figure 26: An implementation of Hoeffding bound(ii) (*Discrete_UCB_Step2.thy*)

5.3.3 Sum of i.i.d. Bounded Variables Compared to Mean Bounded Variables

In section 10.2 we described the variations of Hoeffding-type concentration bounds when applied to the bounded *sum* of i.i.d. random variables compared to the bounded *mean*.

For the mean based UCB the confidence radius shrinks as $1/\sqrt{n}$, making it suitable for problems where the goal is to identify the best arm in terms of per-trial efficiency. Typical use cases include scientific experimentation (e.g., A/B testing) and bandit problems where accuracy of the estimated average reward is the main objective. For the sum based UCB the confidence radius grows like $\sqrt{n}$, making it directly suitable for problems where the total accumulated outcome over time is the optimization target. This formulation emphasizes cumulative performance rather than average efficiency.

5.4 Set Inclusion for Suboptimal Action Selection in UCB

Let $t \in \mathbb{N}$, $a \in A$, and suppose $\omega \in \Omega$, $\Delta(a) = \mu(a^*) - \mu(a) > 0$, i.e., a is suboptimal, $A_{t+1}(t, \omega)$ denotes the action selected by the UCB algorithm at round $t + 1$, $U(t, a, \omega)$ is the UCB index for arm a at time t and $a^* \in A$ is the optimal arm. Then the event that UCB selects a suboptimal arm a at time $t + 1$ is included in the union of two deviation events:

$$\{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \subseteq \{\omega \in \Omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cup \{\omega \in \Omega : \mu(a^*) \leq U(t, a, \omega)\}.$$

Proof Outline.

- (1) $\{\omega : A_{t+1}(t, \omega) = a\} \subseteq \{\omega : \forall b \in A, U(t, a, \omega) \geq U(t, b, \omega)\}$
, by definition of A_{t+1} as the maximizer of $U(t, \cdot, \omega)$.
- (2) $\{\omega : \forall b \in A, U(t, a, \omega) \geq U(t, b, \omega)\} \subseteq \{\omega : U(t, a^*, \omega) \leq U(t, a, \omega)\}$
, because $a^* \in A$.
- (3) $\{\omega : U(t, a^*, \omega) \leq U(t, a, \omega)\} \subseteq \{\omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cup \{\omega : \mu(a^*) \leq U(t, a, \omega)\}$
, holds by basic properties of real numbers (case split).

Conclusion. By transitivity of set inclusion, combining (1), (2), and (3), we obtain the final result:

$$\{\omega : A_{t+1}(t, \omega) = a\} \subseteq \{\omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cup \{\omega : \mu(a^*) \leq U(t, a, \omega)\}.$$

In Isabelle this theorem is defined as shown in the figure below:

```

theorem ucb_suboptimal_bound_set:
  fixes t :: nat                (* current time step *)
  and a :: "'a"                 (* a chosen action *)
  and Δ :: "'a ⇒ real"         (* suboptimality gap function *)
  assumes finite_A: "finite A"   (* action set is finite *)
  and a_in_A: "a ∈ A"           (* action a is in the action set *)
  and a_star_in_A: "a_star ∈ A" (* optimal action is in A *)
  and argmax_exists: "A ≠ {}"
  and subopt_gap: "Δ a > 0"      (* a is suboptimal *)
  and a_not_opt: "∃a'. a' ∈ A ∧ Δ a' > 0" (* there exists a suboptimal action *)
  and delta_a: "Δ a = μ a_star - μ a" (* Δ defined as expected reward difference *)
  and ω_in_Ω: "ω ∈ Ω"          (* ω is in sample space *)
  and asm: "ω ∈ {ω ∈ Ω. A_t_plus_1 t ω = a}" (* ω is such that A_t_plus_1 selects a *)
  and setopt: "∀ω ∈ Ω. ∃a_max ∈ A. ∀b ∈ A. U t b ω ≤ U t a_max ω" (* argmax exists *)
  and A_t_plus_1_maximizes:
    "⋀t ω a. A_t_plus_1 t ω = a ⇒ a ∈ A ∧ (∀b ∈ A. U t a ω ≥ U t b ω)"
  shows "{ω ∈ Ω. A_t_plus_1 t ω = a} ⊆
    {ω ∈ Ω. U t a_star ω ≤ μ a_star} ∪ {ω ∈ Ω. μ a_star ≤ U t a ω}"
proof - [48 lines]
qed

```

Figure 27: A suboptimal bound defined for sets (*Discrete_UCB_Step2.thy*)

5.5 Probability Bound on Suboptimal Action in UCB

Let $t \in \mathbb{N}$, $a \in A$, $\omega \in \Omega$, and assume that $\Delta(a) = \mu(a^*) - \mu(a) > 0$ for some optimal arm $a^* \in A$ and $H = \frac{2\ln(1/\delta)}{(\Delta(a))^2}$. $A_{t+1}(t, \omega)$ is the arm selected by UCB at round $t+1$. Furthermore, $U(t, a, \omega)$ is the UCB index for arm a and $N_n(t, a, \omega)$ is the number of times arm a has been played up to round t . Also, we have that P is a probability space with measurable sets $\mathcal{F}$ and $E = \{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\}$. Finally, let $F = \{\omega \in \Omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\}$ and $G = \{\omega \in \Omega : \mu(a^*) \leq U(t, a, \omega)\} \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\}$.

Then the following probability bound holds:

$$\mathbb{P}(\{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\}) \leq \mathbb{P}(F) + \mathbb{P}(G).$$

Proof. From the previous result, we have the set inclusion:

$$\{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \subseteq \{\omega \in \Omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cup \{\omega \in \Omega : \mu(a^*) \leq U(t, a, \omega)\}.$$

Taking intersection with the event $\{\omega \in \Omega : H \leq N_n(t, a, \omega)\}$, we obtain:

$$\begin{aligned} E &= \{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\} \\ &\subseteq [\{\omega \in \Omega : U(t, a^*, \omega) \leq \mu(a^*)\} \cup \{\omega \in \Omega : \mu(a^*) \leq U(t, a, \omega)\}] \cap \{\omega \in \Omega : H \leq N_n(t, a, \omega)\} \\ &\subseteq F \cup G. \end{aligned}$$

Therefore, by the union bound (5.1),

$$\mathbb{P}(E) \leq \mathbb{P}(F) + \mathbb{P}(G).$$

This is done in Isabelle as follows:

```

theorem ucb_suboptimal_bound_prob_statement:
  fixes t :: nat and a :: 'a and Δ :: "'a ⇒ real"
  assumes finite_A: "finite A" (* action set is finite *)
    and a_star_in_A: "a_star ∈ A" (* optimal action is in A *)
    and argmax_exists: "A ≠ {}"
    and subopt_gap: "Δ a > 0" (* a is suboptimal *)
    and a_not_opt: "∃ a'. a' ∈ A ∧ Δ a' > 0" (* there exists a suboptimal action *)
    and ω_in_Ω: "ω ∈ Ω" (* ω is in sample space *)
    and asm: "ω ∈ {ω ∈ Ω. A_t_plus_1 t ω = a}" (* ω is such that A_t_plus_1 selects a *)
    and setopt: "∀ ω ∈ Ω. ∃ a_max ∈ A. ∀ b ∈ A. U t b ω ≤ U t a_max ω" (* argmax exists *)
    and A_t_plus_1_maximizes:
    "∧ t ω a. A_t_plus_1 t ω = a ⇒ a ∈ A ∧ (∀ b ∈ A. U t a ω ≥ U t b ω)"
  and a_in_A: "a ∈ A"
    and omega_in: "ω ∈ Ω"
    and subopt_gap: "Δ a > 0"
    and delta_a: "Δ a = μ a_star - μ a" (* Δ defined as expected reward difference *)
    and H_def: "H = (2 * ln (1 / δ)) / (Δ a)^2"
    and E_def: "E = {ω ∈ Ω. A_t_plus_1 t ω = a} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)}"
    and F_def: "F = {ω ∈ Ω. U t a_star ω ≤ μ a_star} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)}"
    and G_def: "G = {ω ∈ Ω. μ a_star ≤ U t a ω} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)}"
    and meas_sets: "E ∈ sets M" "F ∈ sets M" "G ∈ sets M"
    and prob_inter: "prob (F ∩ G) ≡ enn2real (emeasure M (F ∩ G))"

  shows "prob ({ω ∈ Ω. A_t_plus_1 t ω = a} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)}) ≤
    prob ({ω ∈ Ω. U t a_star ω ≤ μ a_star} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)}) +
    prob ({ω ∈ Ω. U t a ω ≥ μ a_star} ∩ {ω ∈ Ω. H ≤ real (N_n t a ω)})"
proof - [45 lines]
qed

```

Figure 28: A suboptimal bound for probabilities (*Discrete_UCB_Step2.thy*)

5.5.1 Formalizing Regret Bounds for Discrete UCB

The analysis of regret in multi-armed bandit problems has become a cornerstone of modern online learning theory. Despite the elegance and relative simplicity of the UCB (Upper Confidence Bound) algorithm, a rigorous, mechanized verification of its theoretical guarantees remains an open challenge in the formal methods community. This development contributes toward bridging that gap by providing a formally verified decomposition of suboptimal arm selection events within the Isabelle/HOL proof assistant. The work demonstrates that the logical foundations of UCB-based regret analysis are amenable to formalization, even under the stringent requirements of measure-theoretic probability.

At the heart of this analysis lies the event that a suboptimal arm a is selected at time $t + 1$ and the event that it has been sampled sufficiently many times:

$$\{\omega \in \Omega \mid A_{t+1}(\omega) = a\} \cap \left\{ \omega \in \Omega \mid N_n(t, a, \omega) \geq \frac{2 \ln(1/\delta)}{(\Delta(a))^2} \right\}.$$

This event is of central interest in bounding the probability of selecting suboptimal arms after sufficient exploration has occurred. The key technical insight in the standard UCB proof is that such events can be upper-bounded by the union of two independent failure modes:

$$F := \{\omega \in \Omega \mid U(t, a^*, \omega) \leq \mu(a^*)\} \cap \{\omega \in \Omega \mid H \leq \text{real}(N_n(t, a, \omega))\},$$

$$G := \{\omega \in \Omega \mid \mu(a^*) \leq U(t, a, \omega)\} \cap \{\omega \in \Omega \mid H \leq \text{real}(N_n(t, a, \omega))\},$$

where $H := \frac{2\ln(1/\delta)}{(\Delta(a))^2}$ and $\Delta(a) := \mu(a^*) - \mu(a)$ denotes the suboptimality gap. The correctness of this decomposition is formally verified and shown to be measurable, enabling the application of concentration inequalities.

The empirical mean estimator $M(t, a, \omega)$ and its associated confidence radius are encoded as real-valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The UCB value function is defined by

$$U(t, a, \omega) := M(t, a, \omega) + q \cdot \sqrt{\frac{\ln(1/\delta)}{2 \cdot \text{real}(\max(1, N_n(t, a, \omega)))}}.$$

Within this framework, the proof formalizes the statement that

$$U(t, a^*, \omega) \leq \mu(a^*) \quad \Leftrightarrow \quad M(t, a^*, \omega) - \mu(a^*) \leq -q \cdot \sqrt{\frac{\ln(1/\delta)}{2 \cdot \text{real}(\max(1, N_n(t, a^*, \omega)))}}.$$

This is verified in Isabelle in the lemma defined as shown below

```
lemma U_le_mu_pointwise:
  "U t a_star ω ≤ μ a_star ↔
   M_val t a_star ω - μ a_star ≤
   - q * sqrt (ln (1 / δ) / (2 * real (max 1 (N_n t a_star ω))))"
  unfolding U_def
  using add_le_iff max_Suc_0_eq_1
  by auto
```

Figure 29: Pointwise equivalence (i) (*Discrete_UCB_Step2.thy*)

and analogously,

$$U(t, a, \omega) \geq \mu(a^*) \quad \Leftrightarrow \quad M(t, a, \omega) - \mu(a) \geq \Delta(a) - q \cdot \sqrt{\frac{\ln(1/\delta)}{2 \cdot \text{real}(\max(1, N_n(t, a, \omega)))}}.$$

Consequently, in Isabelle we verified:

```
lemma U_ge_mu_pointwise:
  assumes delta_a: "Δ a = μ a_star - μ a"
  shows
    "U t a ω ≥ μ a_star ↔
     M_val t a ω - μ a ≥ Δ a - q * sqrt (ln (1 / δ) / (2 * real (max 1 (N_n t a ω))))"
  proof - [18 lines]
  qed
```

Figure 30: Pointwise equivalence (ii) (*Discrete_UCB_Step2.thy*)

These expressions mirror the standard confidence interval arguments used in informal proofs, but are embedded here within a fully typed logical system, grounded in Isabelle/HOL's treatment of Lebesgue integration and measurable functions. Crucially, the logical dependencies

are clearly separated: since the relevant tail bounds are now formally established in the library (which we have done generically for i.i.d. bounded variables), the complete probabilistic bound

$$\mathbb{P} \left(A_{t+1} = a \wedge N_n(t, a) \geq \frac{2 \ln(1/\delta)}{(\Delta(a))^2} \right) \leq 2\delta,$$

can be derived without further structural changes.

The strength of this result lies not only in its partial formal verification of a foundational result in online learning, but in establishing that the integration of measure theory, probability, and algorithmic learning is tractable within higher-order logic. This development sets the stage for future formal regret analyses of more complex algorithms — such as contextual bandits, Bayesian UCB, and Thompson sampling where the interaction between probabilistic estimators and decision-making becomes even more intricate.

$$\mathbb{P} \left(\sum_{i=0}^{n-1} Z_i(a) - \mathbb{E} \left[\sum_{i=0}^{n-1} Z_i(a) \right] \geq \epsilon \right) \leq \exp \left(- \frac{2\epsilon^2}{\sum_{i=0}^{n-1} (u-l)^2} \right).$$

These obligations require meticulous unfolding of definitions and precise instantiation of Hoeffding’s inequality lemmas, ensuring consistency between the formal parameters and the instantiated variables.

6 Step 3: Discrete UCB Algorithm Proof Verification

6.1 Fundamental Definitions

This section will conclude our verification of the discrete UCB algorithm in Isabelle following the proof of theorem 15.4 in Rebeschini (2022). In order to verify the theorem we require several assumptions, definitions and the verification of proposition 15.7 (Rebeschini (2022)).

Let us start by defining several properties such as the expectation, probability written as the expectation of indicators and several key expressions in Isabelle.

```

theorem sum_mono:
  fixes f g :: "nat ⇒ real"
  assumes "k ≤ n"
  assumes "∀t ∈ {k..<n}. f t ≤ g t"
  shows "(∑ t = k..<n. f t) ≤ (∑ t = k..<n. g t)"
proof - [15 lines]
qed

definition sample_mean_Z :: "nat ⇒ 'a ⇒ 'b ⇒ real" where
  "sample_mean_Z n a ω = (1 / real n) * (∑ i < n. Z i a ω)"

definition M_fun :: "nat ⇒ 'a ⇒ 'b ⇒ real" where
  "M_fun t a ω = (if N_n (t+1) a ω = 0 then 0
    else (∑ s < t. (if π s ω = a then Z s a ω else 0)) / real (N_n t a ω))"

definition U :: "nat ⇒ 'a ⇒ 'b ⇒ real" where
  "U t a ω = M_fun t a ω + q * sqrt (ln (1 / δ) / (2 * real (max 1 (N_n t a ω))))"

definition A_t_plus_1 :: "nat ⇒ 'b ⇒ 'a" where
  "A_t_plus_1 t ω = (SOME a. a ∈ A ∧ (∀a'. a' ∈ A ⟶ U t a ω ≥ U t a' ω))"

definition prob_eq_Ex :: "'b set ⇒ bool" where
  "prob_eq_Ex E ≡ prob E = expectation (λω. indicator E ω)"

```

Figure 31: Key definitions (*Discrete_UCB_Step3.thy*)

For any measurable set $Y \subseteq \omega$,

$$\text{prob}(Y) = \text{expectation}(\mathbf{1}_Y),$$

where $\mathbf{1}_Y$ is the indicator function of the set Y .

For natural number n , element a , and outcome ω ,

$$\text{sample_mean_Z}(n, a, \omega) = \frac{1}{n} \sum_{i=0}^{n-1} Z(i, a, \omega).$$

For natural number t , element a , and outcome ω ,

$$M(t, a, \omega) = \begin{cases} 0 & \text{if } N_n(t+1, a, \omega) = 0, \\ \frac{\sum_{s=0}^{t-1} (\text{if } \pi(s, \omega) = a \text{ then } Z(s, a, \omega) \text{ else } 0)}{N_n(t, a, \omega)} & \text{otherwise.} \end{cases}$$

$$U(t, a, \omega) = M(t, a, \omega) + q \sqrt{\frac{\ln(1/\delta)}{2 \max(1, N_n(t, a, \omega))}}.$$

Finally,

$$A_{t+1}(t, \omega) = \text{some } a \in A \text{ such that } \forall a' \in A, \quad U(t, a, \omega) \geq U(t, a', \omega).$$

Fix $a \in A$, $t \in \mathbb{N}$, $\omega \in \Omega$, and $\Delta : A \rightarrow \mathbb{R}$.

6.2 Proposition 15.7

Assume the hypotheses:

- i) $a \in A$, $\mathbb{P}$ is a probability space, $\omega \in \Omega$,
- ii) $\Delta(a) > 0$, $\exists a' \in A, \Delta(a') > 0$, $\forall a \in A, \Delta(a) = \mu_{a^*} - \mu_a$,
- iii) $\forall a \in A$, $\mathbb{E}[N_n(n, a)] \leq s(n) + \mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge s(t) \leq N_n(t, a)\}} \right]$,
- iv) $\forall a \in A, \forall t \in \{k, \dots, n-1\}$, $\mathbb{P}(A_{t+1} = a \cap \{N_n(t, a) \geq \frac{2 \ln(1/\delta)}{\Delta(a)^2}\}) \leq 2\delta$,
- v) $\varepsilon > 0$, $\forall t \in \{k, \dots, n-1\}$, $t > 0$, $\delta = \frac{1}{t^\varepsilon}$,
- vi) $u \in \mathbb{N}, s(u) = \frac{2\varepsilon \ln u}{\Delta(a)^2}$,
 $\forall a \in A, \forall t \in \{k, \dots, n-1\}$, $\{\pi(t+1) = a \wedge s(t) \leq N_n(t, a)\} = \{A_{t+1} = a\} \cap \left\{ N_n(t, a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2} \right\}$,
- vii) $\mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge s(t) \leq N_n(t, a)\}} \right] = \sum_{t=k}^{n-1} \mathbb{P}(\pi(t+1) = a \wedge s(t) \leq N_n(t, a))$.

Please note that assumptions (iii) and (iv) follow from verifying the corresponding expressions for an arbitrary fixed value of $a \in A$.

Specifically, Section 4.6 establishes that for any $a \in A$,

$$\mathbb{E}[N_n(n, a)] \leq s(n) + \mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge s(t) \leq N_n(t, a)\}} \right].$$

Moreover, based on the lower and upper tail bound results from Hoeffdings lemma combined with the probability bound on sub-optimal actions (Section 5) we can conclude that for any $a \in A$ and appropriate H ,

$$\mathbb{P}(\{\omega \mid A_{t+1}(t, \omega) = a\} \cap \{\omega \mid N_n(t, a, \omega) \geq H\}) \leq 2\delta.$$

These assumptions provide important connections between the events defined by the policy's action selections and the counts of those actions, facilitating precise probabilistic analysis.

The assumption `eq_sets_optimum` (assumption (vi)) states that for every action $a \in A$ and time step $t \in \{k, \dots, n-1\}$, the event

$$\{\omega \mid \pi(t+1, \omega) = a \wedge N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{(\Delta a)^2}\},$$

is exactly equal to the intersection

$$\{\omega \in \Omega \mid A_{t+1}(t, \omega) = a\} \cap \{\omega \in \Omega \mid N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{(\Delta a)^2}\}.$$

This equality ensures that the event described by the policy choosing action a under the threshold condition corresponds precisely to a measurable intersection of events, enabling rigorous probability computations. Note that, we use the general definition that for $t \in \mathbb{N}$, $s(t) = \frac{2\varepsilon \ln t}{\Delta(a)^2}$

The assumption `assms_lin_expect` (assumption (vii)) expresses the linearity of expectation for these indicator events, stating that

$$\mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge N_n(t,a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\}} \right] = \sum_{t=k}^{n-1} \mathbb{E} \left[\mathbf{1}_{\{\pi(t+1)=a \wedge N_n(t,a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\}} \right].$$

This allows interchanging summation and expectation over indicator functions, which is fundamental in analyzing the expected count of action selections satisfying the threshold criteria.

In the proof we express $s(n)$ explicitly

$$s(n) = \frac{2\varepsilon \ln n}{\Delta(a)^2}.$$

Then we define the bound $\mathbb{E}[N_n(n, a)]$:

$$\mathbb{E}[N_n(n, a)] \leq s(n) + \mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge s(t) \leq N_n(t,a)\}} \right].$$

By linearity of expectation and the indicator-probability relation, this is

$$= s(n) + \sum_{t=k}^{n-1} \mathbb{P}(\pi(t+1) = a \wedge s(t) \leq N_n(t, a)).$$

Using the set equality from assumptions,

$$= s(n) + \sum_{t=k}^{n-1} \mathbb{P} \left(A_{t+1} = a \cap \left\{ N_n(t, a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2} \right\} \right).$$

We then apply the probability bound. From the assumption,

$$\mathbb{P} \left(A_{t+1} = a \cap \left\{ N_n(t, a) \geq \frac{2 \ln(1/\delta)}{\Delta(a)^2} \right\} \right) \leq 2\delta,$$

and with $\delta = t^{-\varepsilon}$, we get

$$\mathbb{P} \left(A_{t+1} = a \cap \left\{ N_n(t, a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2} \right\} \right) \leq \frac{2}{t^\varepsilon}.$$

Note that in order to use substitution with the choice $\delta = t^{-\varepsilon}$ we verified in Isabelle a key result (Generalized Bound) that states:

$$\forall a \in A, \forall t \in \{k, \dots, n-1\}, \quad \mathbb{P} \left(\{\omega : A_{t+1}(t, \omega) = a\} \cap \{\omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\} \right) \leq \frac{2}{t^\varepsilon}.$$

Also, we sum over t

$$\sum_{t=k}^{n-1} P \left(A_{t+1} = a \cap \left\{ N_n(t, a) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2} \right\} \right) \leq \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon}.$$

by verifying first

$$(\text{Sum Monotonicity}) \quad \sum_{t=k}^{n-1} \mathbb{P} \left(\{\omega : A_{t+1}(t, \omega) = a\} \cap \{\omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\} \right) \leq \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon}.$$

Note that, we require the verification of a generalized bound and sum monotonicity. The steps of the proofs in Isabelle are shown in Appendices A4-A5 (10.3 - 10.4).

Combining all,

$$\forall a \in A, \mathbb{E}[N_n(n, a)] \leq \frac{2\varepsilon \ln n}{\Delta(a)^2} + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon}.$$

This concludes the proof. The proposition is defined in Isabelle as shown in figure 32 below:

```

theorem proposition_15_7:
  assumes
    a_in_A: "a ∈ A"
  and "ω ∈ Ω"
  and subopt_gap: "Δ a > 0"
  and a_not_opt: "∃ a'. Δ a' > 0"
  and delta_a: "∀ a ∈ A. Δ a = μ a_star - μ a"
  and "k ≤ n"
  and from_UCB_step1: "∀ a ∈ A. expectation (λω. real (N_n n a ω)) ≤
    s n + expectation (λω. (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N_n t a ω) then 1 else 0))"
  and from_UCB_step2: "∀ a ∈ A. ∀ t ∈ {k..<n}. prob {ω ∈ Ω. A_t_plus_1 t ω = a} ∩
    {ω ∈ Ω. N_n t a ω ≥ (2 * ln (1 / δ)) / (Δ a)^2} ≤ 2 * δ"
  and eps_pos: "ε > 0"
  and t_gt0: "∀ t ∈ {k..<n}. t > 0"
  and choice_delta: "∀ t ∈ {k..<n}. δ = 1 / (real t power ε)"
  and s_form: "∀ a ∈ A. ∀ u. s u = (2 * ε * ln (real u)) / ((Δ a)^2)"
  and subset_meas: "∀ a ∈ A. ∀ t ∈ {k..<n}. ∀ ω ∈ Ω. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ⊆
    prob {ω. π (Suc t) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ∩ space M"
  and finiteness: "∀ t ∈ {k..<n}. ∀ a ∈ A. emeasure M {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} < ∞"
  and measurable_set: "∀ t ∈ {k..<n}. ∀ a ∈ A. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ∈ sets M"

  and eq_sets_optimum:
    "∀ a ∈ A. ∀ t ∈ {k..<n}. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} =
      {ω ∈ Ω. A_t_plus_1 t ω = a} ∩ {ω ∈ Ω. N_n t a ω ≥ (2 * ε * ln (real t)) / (Δ a)^2}"

  shows
    "∀ a ∈ A. expectation (λω. real (N_n n a ω)) ≤ s n + (∑ t = k..<n. 2 / (real t power ε))"

proof - [240 lines]

qed

```

Figure 32: Verification of proposition 15.7 (*Discrete_UCB_Step3.thy*)

6.3 Theorem 15.4

We assume the following:

- (i) $a \in A$, A is finite, P is a probability space with finite measure,
- (ii) $\forall a \in A$. the random variable $\omega \mapsto N_n(n, a, \omega)$ is integrable,
- (iii) $\forall a \in A$. $\Delta(a) > 0$, $\Delta(a) = \mu(a_*) - \mu(a)$,
- (iv) $k \leq n$, $\sum_{a \in A} N_n(n, a, \omega) = n$ (pointwise),
- (v) $\mathbb{E}[R_n(n, \omega)] = \sum_{a \in A} \Delta(a) \mathbb{E}[N_n(n, a, \cdot)]$,
- (vi) $\forall a \in A : \mathbb{E}[N_n(n, a)] \leq s(n) + \mathbb{E} \left[\sum_{t=k}^{n-1} \mathbf{1}_{\{\pi(t+1)=a \wedge s(t) \leq N_n(t, a)\}} \right]$,
- (vii) (Proposition 15.7) $\forall a \in A : \mathbb{E}[N_n(n, a)] \leq s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon}$,
- (viii) $s(u) = \frac{2\varepsilon \ln u}{\Delta(a)^2}$ for each $a \in A$.

These assumptions and of proposition 15.7 are combined for the theorem verification and written in Isabelle as follows:

```

theorem theorem_15_4:
  assumes a_in_A: "a ∈ A"
  and "finite A" and "∀a ∈ A. integrable M (λω. real (N_n n a ω))"
  and ω_in_Ω: "ω ∈ Ω"
  and subopt_gap: "∀a ∈ A. Δ a > 0"
  and a_not_opt: "∃ a' ∈ A. Δ a' > 0"
  and delta_a: "∀a ∈ A. Δ a = μ a_star - μ a" and "k ≤ n" and n_count_assm_pointwise: "(∑a∈A. real (N_n n a ω))"
  and expected_regret_prop_15_1: "expectation (λω. R_n n ω) = (∑a∈A. Δ a * expectation (λω. real (N_n n a ω)))"
  and from_UCB_step1: "∀a ∈ A. expectation (λω. real (N_n n a ω)) ≤ s n + expectation (λω. (∑ t = k..<n. if π (t+1) ω = a ∧ s t ≤ real (N_n t a ω) then 1 else 0))"
  and from_UCB_step2: "∀a ∈ A. ∀t ∈ {k..<n}. prob {ω ∈ Ω. A t plus 1 t ω = a} ∩ {ω ∈ Ω. N_n t a ω ≥ (2 * ln (1 / δ)) / (Δ a)^2} ≤ 2 * δ"
  and eps_pos: "ε > 0" and t_gt0: "∀t ∈ {k..<n}. t > 0" and choice_delta: "∀t ∈ {k..<n}. δ = 1 / (real t powr ε)"
  and s_form: "∀a ∈ A. ∀u. s u = (2 * ε * ln (real u)) / ((Δ a)^2)"
  and subset_meas: "∀a ∈ A. ∀t ∈ {k..<n}. ∀ω ∈ Ω. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ⊆ {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ∩ space M"
  and prob_eq_E_assm: "∀a ∈ A. ∀t ∈ {k..<n}. prob {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ∩ space M"
  and finiteness: "∀t ∈ {k..<n}. ∀a ∈ A. emeasure M {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} < ∞"
  and measurable_set: "∀t ∈ {k..<n}. ∀a ∈ A. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} ∈ sets M"
  and eq_sets_optimum:
    "∀a ∈ A. ∀t ∈ {k..<n}. {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)} = {ω ∈ Ω. A t plus 1 t ω = a} ∩ {ω ∈ Ω. N_n t a ω ≥ (2 * ε * ln (real t)) / (Δ a)^2}"
  and assms_lin_expect: "∀a ∈ A. expectation (λω. ∑ t = k..<n. (if π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω) then 1 else 0)) = (∑ t = k..<n. expectation (λω. indicat_real {ω. π (t+1) ω = a ∧ 2 * ε * ln (real t) / (Δ a)^2 ≤ real (N_n t a ω)}))"
  and mono_sum_sets:
    " (∀a ∈ A. Δ a * expectation (λω. real (N_n n a ω)) ≤ Δ a * (s n + (∑ t = k..<n. 2 / (real t powr ε))))
    ⇒ (∑a ∈ A. Δ a * expectation (λω. real (N_n n a ω))) ≤ (∑a ∈ A. Δ a * (s n + (∑ t = k..<n. 2 / (real t powr ε))))"
  shows "expectation (λω. R_n n ω) ≤ (∑a ∈ A. Δ a * ((2 * ε * ln (real n)) / ((Δ a)^2) + (∑ t = k..<n. 2 / (real t powr ε))))"
  proof - [43 lines]

```

Figure 33: Theorem 15.4 (*Discrete_UCB_Step3.thy*)

We describe the proof steps exactly as in our formal development.

By the given expected-regret decomposition (assumption),

$$\mathbb{E}[R_n(n, \omega)] = \sum_{a \in A} \Delta(a) \mathbb{E}[N_n(n, a, \cdot)].$$

By Proposition 15.7, we have the uniform bound for every arm $a \in A$:

$$\mathbb{E}[N_n(n, a)] \leq s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon}.$$

Fix arbitrarily $a \in A$ then with the strict positivity $\Delta(a) > 0$ we obtain

$$\Delta(a) \mathbb{E}[N_n(n, a)] \leq \Delta(a) \left(s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon} \right).$$

The inequality is preserved upon multiplication because $\Delta(a) > 0$.

Thus we have the pointwise inequality

$$\forall a \in A : \quad \Delta(a) \mathbb{E}[N_n(n, a)] \leq \Delta(a) \left(s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon} \right).$$

Applying the monotonicity for finite sums (In Isabelle our `mono_sum_sets` hypothesis), which lifts a pointwise inequality on each summand to an inequality on the sum, we obtain

$$\sum_{a \in A} \Delta(a) \mathbb{E}[N_n(n, a)] \leq \sum_{a \in A} \Delta(a) \left(s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon} \right).$$

By substitution:

$$\mathbb{E}[R_n(n, \omega)] \leq \sum_{a \in A} \Delta(a) \left(s(n) + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon} \right).$$

By the assumed form of s (the hypothesis $s(u) = \frac{2\varepsilon \ln u}{\Delta(a)^2}$ for each a), we replace $s(n)$ to obtain the advertised bound:

$$\boxed{\mathbb{E}[R_n(n, \omega)] \leq \sum_{a \in A} \Delta(a) \left(\frac{2\varepsilon \ln n}{\Delta(a)^2} + \sum_{t=k}^{n-1} \frac{2}{t^\varepsilon} \right)}.$$

This completes the proof.

7 Continuous Bandit Algorithms

This paper also presents an initial exploration into extending the Upper Confidence Bound (UCB) algorithm to a continuous-time multi-armed bandit setting, where rewards follow stochastic differential equations with volatility modeled by Brownian motion. In appendix (section 10.5) we formulate the regret bound in this continuous framework by leveraging martingale concentration inequalities and Itô calculus, which generalize classical discrete-time concentration results.

8 Conclusion

This thesis has presented a machine-checked formalization of regret analysis in stochastic multi-armed bandit problems, culminating in a complete verification of the Upper Confidence Bound (UCB) regret guarantee within Isabelle/HOL. We began by constructing a rigorous probabilistic framework in which all key objects: arms, rewards, optimal actions, and suboptimality gaps were formally specified and their mathematical properties encoded. Within this setting we established a precise correspondence between informal definitions from the literature and their formally verified counterparts, showing that cumulative regret admits a decomposition into weighted arm-pull counts that extends naturally to expectations.

The verification of the UCB analysis proceeded in three steps. First, we proved set-theoretic lemmas enabling systematic decomposition of arm counts, and established the equivalence between cardinalities and indicator sums, leading to an expectation inequality for suboptimal pulls. Second, we formalized the UCB machinery itself, together with a suite of probability-theoretic results, including a union-bound decomposition of selection events and Hoeffding-type inequalities, yielding high-probability control of suboptimal choices. Finally, combining these ingredients, we derived the canonical logarithmic sample-complexity and regret bounds.

This work demonstrates both the fidelity of Isabelle/HOL in formalizing the structure of bandit regret analysis and the feasibility of machine-checked verification of central results in learning theory. Future work could extend this framework to more general bandit models and also complex continuous-time algorithms.

9 References

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10 Appendix

10.1 A0: Locale Definition Bandit policy

```

locale bandit_policy = bandit_problem + prob_space +
  fixes  $\Omega$  :: "'b set"
  and  $\mathcal{F}$  :: "'b set set"
  and  $\pi$  :: "nat  $\Rightarrow$  'b  $\Rightarrow$  'a"
  and  $N\_n$  :: "nat  $\Rightarrow$  'a  $\Rightarrow$  'b  $\Rightarrow$  nat"
  assumes measurable_policy: " $\forall t. \pi\ t \in \text{measurable } M\ (\text{count\_space } A)$ "
  and  $N\_n$  def: " $\forall n\ a\ \omega. N\_n\ n\ a\ \omega = \text{card } \{t \in \{0..<n\}. \pi\ (t+1)\ \omega = a\}$ "
  and count_assm_pointwise: " $\forall n\ \omega. (\sum a \in A. \text{real } (N\_n\ n\ a\ \omega)) = \text{real } n$ "
begin

```

Figure 34: Locale (*MSc_Project_Discrete_Prob15-1.thy*)

10.2 A1: Hoeffding Bound

10.2.1 On the Placement of n in Hoeffding Bounds

Consider n independent random variables $Z_0, Z_1, \dots, Z_{n-1}$ such that

$$l \leq Z_j \leq u \quad \text{almost surely for all } j = 0, \dots, n-1,$$

and define the sum

$$S_n := \sum_{j=0}^{n-1} Z_j, \quad \text{with mean } \mu := \mathbb{E}[S_n] = \sum_{j=0}^{n-1} \mathbb{E}[Z_j].$$

Hoeffding's inequality states that for any $\varepsilon > 0$,

$$\Pr(S_n \leq \mu - \varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{j=0}^{n-1} (u-l)^2}\right) = \exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right).$$

If we now set the right-hand side equal to a confidence level $\delta \in (0, 1)$, i.e.,

$$\exp\left(-\frac{2\varepsilon^2}{n(u-l)^2}\right) = \delta,$$

then solving for ε gives

$$\varepsilon = |u-l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}.$$

Notice that in this form, ε scales as $\sqrt{n}$ because the inequality bounds the *sum* S_n . That is, larger n naturally allows for a larger deviation from the mean in absolute terms.

When bounding the **sum** S_n , $\varepsilon \sim \sqrt{n}$, i.e., the deviation grows with the number of variables:

$$\varepsilon_{\text{sum}} = |u-l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}.$$

When bounding the **mean** $\bar{Z}_n$, $\epsilon \sim 1/\sqrt{n}$, i.e., the deviation decreases as the sample size increases:

$$\epsilon_{\text{mean}} = |u - l| \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

In other words, moving n from the numerator (sum case) to the denominator (mean case) corresponds precisely to normalizing by n and reflects the law of large numbers: the more samples you average, the smaller the expected deviation of the mean from its expectation.

In practice, for UCB-style bounds, the relevant quantity is usually the *sample mean*, so n appears in the denominator, yielding a shrinking confidence interval as more observations are collected. Using n in the numerator, as in $\epsilon = |u - l| \sqrt{\frac{n}{2} \ln \frac{1}{\delta}}$, is valid only when bounding the sum rather than the mean.

In traditional Upper Confidence Bound (UCB) algorithms, the focus is on estimating the *expected mean reward* of each arm. The canonical mean-based UCB for arm i at time t is:

$$\text{UCB}_i^{\text{mean}}(t) = \hat{\mu}_i(t) + \sqrt{\frac{2 \ln t}{n_i(t)}}, \quad (1)$$

where $\hat{\mu}_i(t) = \frac{1}{n_i(t)} \sum_{j=1}^{n_i(t)} X_{i,j}$ is the sample mean, and $n_i(t)$ is the number of pulls of arm i . The confidence interval shrinks with $1/\sqrt{n_i(t)}$, reflecting increasing certainty in the mean as more observations are collected. This formulation is appropriate when the primary goal is to *identify the arm with the highest per-pull expected reward*.

In contrast, a *sum-based UCB* bounds the cumulative reward $S_i(t) = \sum_{j=1}^{n_i(t)} X_{i,j}$. Applying Hoeffding's inequality to the sum yields:

$$\Pr \left(|S_i(t) - \mathbb{E}[S_i(t)]| \geq \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{n_i(t)(b-a)^2} \right), \quad (2)$$

for rewards bounded in $[a, b]$. Solving for ε at confidence level δ gives the sum-based confidence interval:

$$\varepsilon_i(t) = (b-a) \sqrt{\frac{n_i(t)}{2} \ln \frac{1}{\delta}}. \quad (3)$$

The sum-based UCB is then defined as:

$$\text{UCB}_i^{\text{sum}}(t) = S_i(t) + (b-a) \sqrt{\frac{n_i(t)}{2} \ln \frac{1}{\delta}}. \quad (4)$$

Unlike mean-based UCB, the confidence interval *grows with* $\sqrt{n_i(t)}$ rather than shrinking. This design favors arms that already have high cumulative reward and/or have been pulled

frequently, making it appropriate when the objective is to *maximize total accumulated reward* rather than the per-pull mean.

10.3 A4: Sum Monotonicity (Discrete_UCB_Step3.thy)

Fix functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ and integers $k \leq n$. Assume

$$\forall t \in \{k, \dots, n-1\}, \quad f(t) \leq g(t).$$

Then,

$$\sum_{t=k}^{n-1} f(t) \leq \sum_{t=k}^{n-1} g(t).$$

First, observe that for all $t \in \{k, \dots, n-1\}$, the difference

$$g(t) - f(t) \geq 0,$$

by the assumption.

Therefore,

$$\sum_{t=k}^{n-1} (g(t) - f(t)) \geq 0.$$

Using the linearity of summation, we rewrite this as

$$\sum_{t=k}^{n-1} g(t) - \sum_{t=k}^{n-1} f(t) \geq 0.$$

Rearranging the inequality, we obtain

$$\sum_{t=k}^{n-1} f(t) \leq \sum_{t=k}^{n-1} g(t),$$

which completes the proof.

10.4 A5: Generalized Bound Discrete_UCB_Step3.thy

Let $a \in A$, $k \leq t < n$, and $\varepsilon > 0$. Then

$$\mathbb{P}\left(\{\omega \in \Omega : A_{t+1}(t, \omega) = a\} \cap \left\{\omega \in \Omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\right\}\right) \leq \frac{2}{t^\varepsilon}.$$

Fix $a \in A$.

Let $t \in \{k, \dots, n-1\}$. Define $\delta = \frac{1}{t^\varepsilon}$.

From the given assumption (or previously established bound) we have

$$\mathbb{P}\left(\{\omega : A_{t+1}(t, \omega) = a\} \cap \left\{\omega : N_n(t, a, \omega) \geq \frac{2 \ln(1/\delta)}{\Delta(a)^2}\right\}\right) \leq 2\delta.$$

Note that

$$\ln \frac{1}{\delta} = \ln(t^\varepsilon) = \varepsilon \ln t,$$

so the threshold can be rewritten as

$$\frac{2 \ln(1/\delta)}{\Delta(a)^2} = \frac{2\varepsilon \ln t}{\Delta(a)^2}.$$

Hence, the event sets satisfy

$$\left\{\omega : N_n(t, a, \omega) \geq \frac{2 \ln(1/\delta)}{\Delta(a)^2}\right\} = \left\{\omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\right\}.$$

By substituting this equivalence back into the probability bound, we get

$$\mathbb{P}\left(\{\omega : A_{t+1}(t, \omega) = a\} \cap \left\{\omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\right\}\right) \leq 2\delta.$$

Recalling the definition of δ , we have

$$2\delta = 2 \cdot \frac{1}{t^\varepsilon} = \frac{2}{t^\varepsilon}.$$

Therefore,

$$\mathbb{P}\left(\{\omega : A_{t+1}(t, \omega) = a\} \cap \left\{\omega : N_n(t, a, \omega) \geq \frac{2\varepsilon \ln t}{\Delta(a)^2}\right\}\right) \leq \frac{2}{t^\varepsilon},$$

which completes the proof.

10.5 A7. Continuous Time UCB Algorithm: An Introductory Exploration

The Upper Confidence Bound (UCB) algorithm's discrete-time version is well studied; for volatility-aware decision making, continuous-time extension remains an emerging and highly relevant field. We can define the regret bound for a continuous-time UCB algorithm, based on stochastic differential equations (SDEs) and martingale concentration inequalities.

Consider a continuous-time multi-armed bandit problem with K arms, where each arm i follows a stochastic reward process governed by the stochastic differential equation (SDE):

$$dX_i(t) = \mu_i dt + \sigma_i dW_i(t), \tag{5}$$

where $X_i(t)$ is the stochastic reward process, μ_i is the true mean reward, σ_i represents volatility, and $W_i(t)$ is a standard Brownian motion.

At any time t , the decision-maker selects an arm $A(t)$ and receives a reward accordingly. The objective is to minimize the cumulative regret:

$$R(T) = \mathbb{E} \left[\int_0^T (\mu^* - \mu_{A(t)}) dt \right], \quad (6)$$

where $\mu^* = \max_i \mu_i$ is the optimal mean reward.

The empirical mean reward for arm i up to time t is given by:

$$\hat{\mu}_i(t) = \frac{1}{T_i(t)} \int_0^t \mathbf{1}_{\{A(s)=i\}} dX_i(s), \quad (7)$$

where $T_i(t)$ is the total time spent on arm i :

$$T_i(t) = \int_0^t \mathbf{1}_{\{A(s)=i\}} ds. \quad (8)$$

Using Itô's lemma and linearity of expectation:

$$\mathbb{E}[\hat{\mu}_i(t)] = \mu_i. \quad (9)$$

Define the martingale using Isabelle's probability library Nipkow et al. (2024):

$$M_i(t) = \int_0^t \mathbf{1}_{\{A(s)=i\}} \sigma_i dW_i(s). \quad (10)$$

Applying Doob's martingale inequality Doob (1953) and Dudley's maximal inequality Dudley (1967), we obtain the concentration bound :

$$\mathbb{P}(|\hat{\mu}_i(t) - \mu_i| \geq \epsilon) \leq 2 \exp \left(-\frac{T_i(t)\epsilon^2}{2\sigma_i^2} \right). \quad (11)$$

To show this we note that the martingale $M_i(t)$ is defined as:

$$M_i(t) = \int_0^t \mathbf{1}_{\{A(s)=i\}} \sigma_i dW_i(s). \quad (12)$$

with variance:

$$\mathbb{E}[|M_i(T)|^2] = \sigma_i^2 \int_0^T \mathbb{E}[\mathbf{1}_{\{A(s)=i\}}] ds. \quad (13)$$

Note that, if we assume

$$p_i = \mathbb{E}[\mathbf{1}_{\{A(s)=i\}}], \quad (14)$$

we get:

$$\mathbb{E}[|M_i(T)|^2] = \sigma_i^2 p_i T. \quad (15)$$

Doob's maximal inequality states:

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |M_i(t)| \right] \leq 2 \mathbb{E}[|M_i(T)|]. \quad (16)$$

Using Markov's inequality:

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} |M_i(t)| \geq \epsilon \right) \leq \frac{1}{\epsilon} \mathbb{E} \left[\sup_{0 \leq t \leq T} |M_i(t)| \right]. \quad (17)$$

Substituting Doob's inequality:

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} |M_i(t)| \geq \epsilon \right) \leq \frac{2}{\epsilon} \mathbb{E}[|M_i(T)|]. \quad (18)$$

Using Itô's isometry:

$$\mathbb{E}[|M_i(T)|^2] = \mathbb{E} \left[\left(\int_0^T \mathbf{1}_{\{A(s)=i\}} \sigma_i dW_i(s) \right)^2 \right]. \quad (19)$$

Applying the isometry property:

$$\mathbb{E}[|M_i(T)|^2] = \sigma_i^2 \int_0^T \mathbb{E}[\mathbf{1}_{\{A(s)=i\}}] ds. \quad (20)$$

We assume $\mathbb{E}[\mathbf{1}_{\{A(s)=i\}}] = p_i$ therefore we obtain $\mathbb{E}[|M_i(T)|^2] = \sigma_i^2 p_i T$ (see equation 35). Thus,

$$\mathbb{E}[|M_i(T)|] = \sqrt{\mathbb{E}[|M_i(T)|^2]} = \sigma_i \sqrt{p_i T}. \quad (21)$$

$$\mathbb{P}(|\hat{\mu}_i(t) - \mu_i| \geq \epsilon) \leq \frac{2\sigma_i \sqrt{p_i T}}{\epsilon}. \quad (22)$$

Applying a concentration inequality (Dudley's inequality), we obtain:

$$\mathbb{P}(|\hat{\mu}_i(t) - \mu_i| \geq \epsilon) \leq 2 \exp \left(-\frac{T_i(t) \epsilon^2}{2\sigma_i^2} \right). \quad (23)$$

Using standard techniques from concentration inequalities, we set $\epsilon = \sqrt{\frac{\beta \log t}{T_i(t)}}$, with $\beta = 2\sigma_i^2$, leading to the definition of the Upper Confidence Bound (UCB):

$$U_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{\beta \log t}{T_i(t)}}. \quad (24)$$

At each time t , the algorithm selects the arm with the highest confidence bound:

$$A(t) = \arg \max_i U_i(t). \quad (25)$$

Define the suboptimality gap for arm i as $\Delta_i = \mu^* - \mu_i$. An arm i is chosen when:

$$U_i(t) \geq U^*(t). \quad (26)$$

Expanding and rearranging terms:

$$\mu_i - \mu^* + 2\sqrt{\frac{\beta \log t}{T_i(t)}} \geq 0. \quad (27)$$

Solving for $T_i(t)$:

$$T_i(t) \leq \frac{4\beta \log t}{\Delta_i^2}. \quad (28)$$

The total regret is bounded using the classical analysis of bandit algorithms Bubeck and Cesa-Bianchi (2012):

$$R(T) \leq \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(T)] = O\left(\sum_{i:\Delta_i>0} \frac{\log T}{\Delta_i}\right). \quad (29)$$

Thus, the regret of continuous-time UCB scales as:

$$R(T) = O(\log T). \quad (30)$$